

Retrospective: DPW Through the Years

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Boeing Technical Fellow, Retired



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- **Introduction**
- **Brief Overview of DPW1, DPW2, and DPW3**
- **Introduction of the NASA High Speed Common Research Model**
- **Brief Introduction to DPW4, DPW5, Aeroelastics, DPW6, and DPW7**
- **Grid Convergence Results**
- **Angle-of-Attack Sweeps**
- **Other Results**
- **Issues**
- **General Observations and Comments**

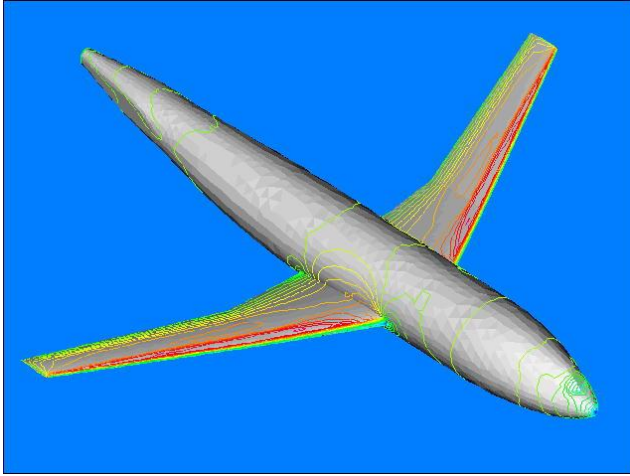
Objectives

- To assess state-of-the-art CFD methods as practical aerodynamic tools for prediction of forces & moments on industry-relevant geometries, with a focus on drag
- To provide an impartial international forum for evaluating the effectiveness of CFD NS solvers
- To identify areas needing additional research and development

Principles

- Use public-domain subject geometries and test data
- Maintain a public-domain accessible database of geometries, grids, and results

DLR-4F Wind Tunnel Model



Required:

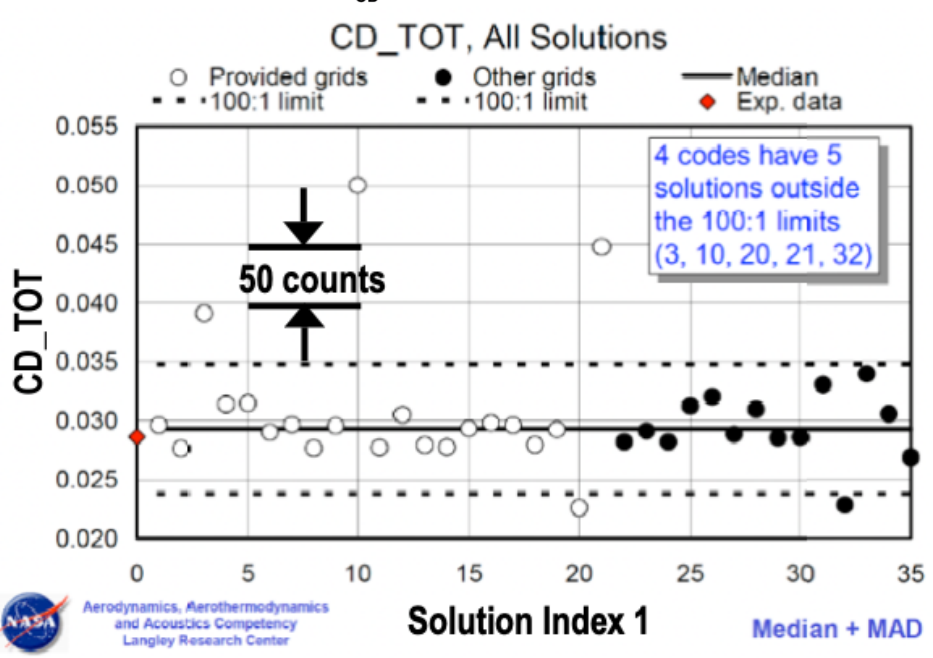
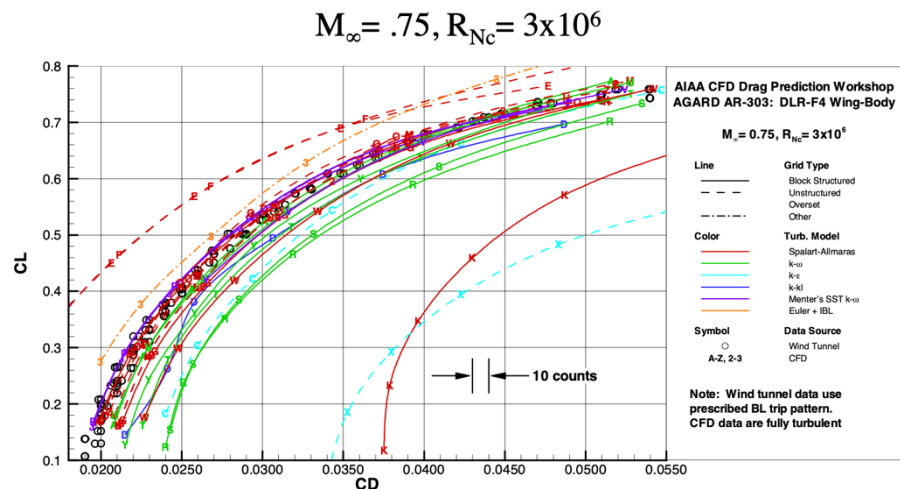
- **Case 1:** $M_\infty = 0.75$, $C_L = 0.500 \pm 0.005$
- **Case 2:** $M_\infty = 0.75$, $\alpha = -3^\circ, -2^\circ, -1^\circ, 0^\circ, 1^\circ, 2^\circ$

Optional:

- **Cases 3:** M_∞ sweep, $C_L = .500$
- **Cases 4:** M_∞ sweep, $C_L = .400, .600$

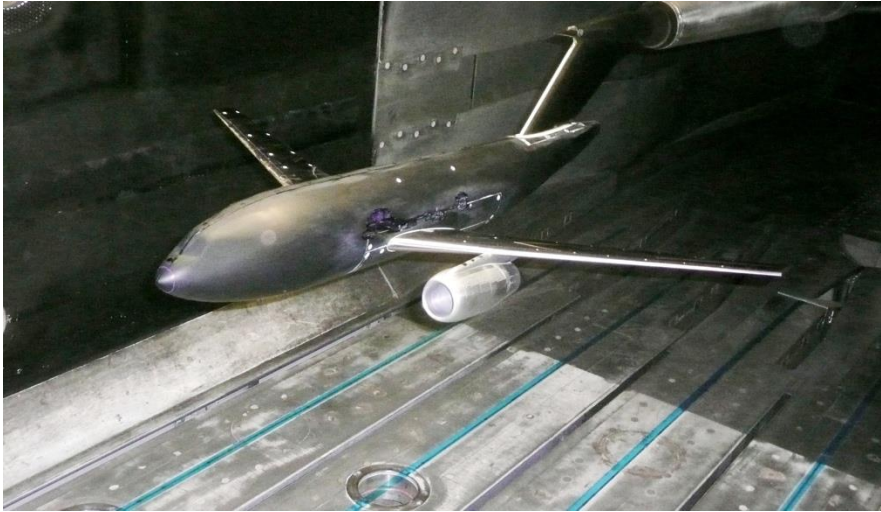
Standard Grids Supplied, Custom Grids Encouraged

- **18 Participants**
- **28 Submittals**
- **14 Codes**
- **Grids Types**
 - 8 Block Structured
 - 2 Overset
 - 7 Unstructured
 - 1 Cartesian
- **Turbulence Models**
 - 14 Spalart-Allmaras (all variations)
 - 10 k-omega
 - 2 Realizable k- epsilon
 - Other



- Scatter amongst CFD solutions at cruise significantly higher than anyone anticipated
 - Pockets of separated flow a leading cause
- Hard to isolate and quantify effects causing variability
 - Turb model, grid, code, transition, geometry (some simplified)
- Problematic grids - okay for one code, not another
- Comparison with experiment became a much larger focus that anticipated
 - CFD C_L generally too high at a given a
 - CFD C_M generally more nose-down

DLR-6F Wind Tunnel Model



Required: Both Wing-Body and Wing-Body-Strut Nacelle

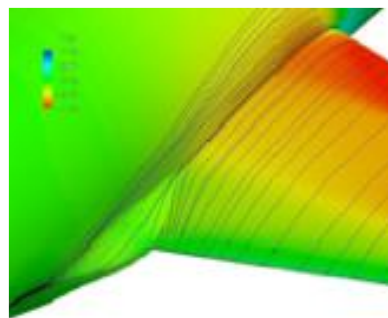
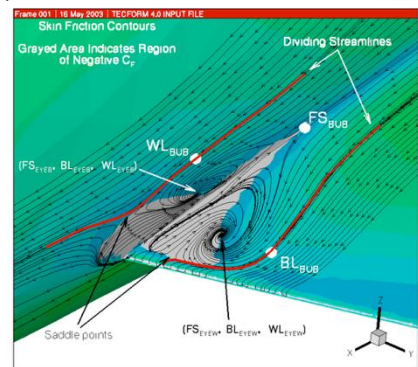
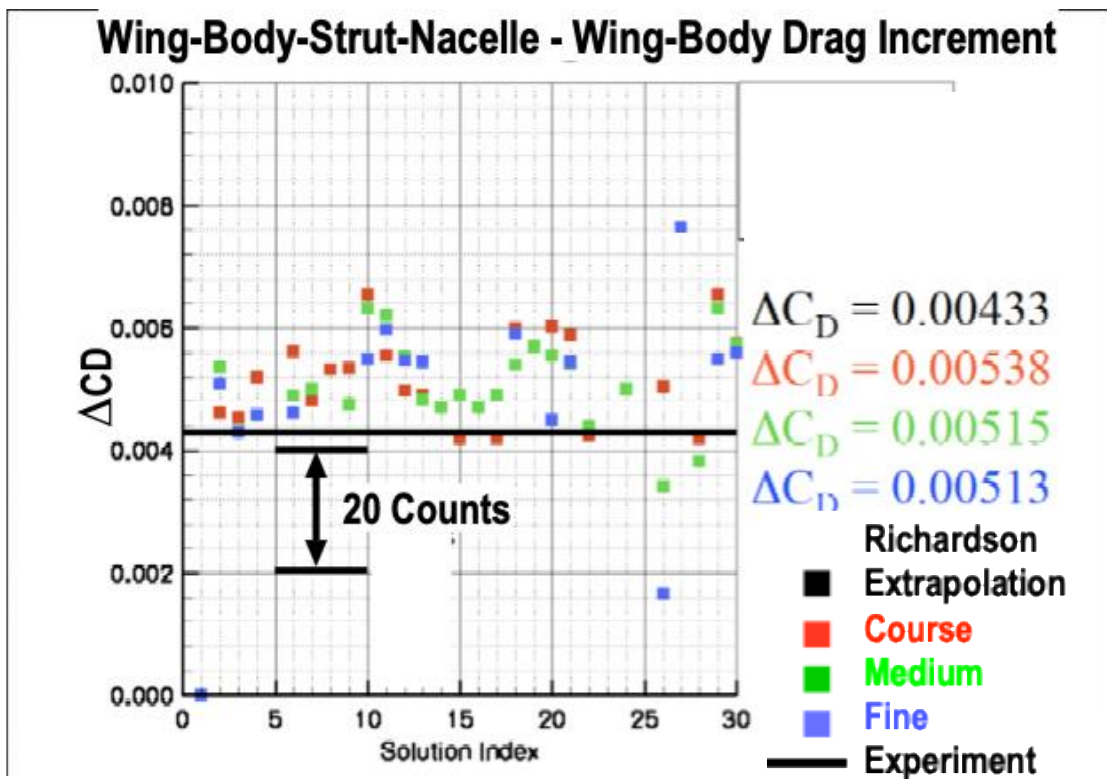
- **Case 1:** $M_\infty = 0.75$, $C_L = 0.500 \pm 0.005$
- **Case 2:** $M_\infty = 0.75$, $\alpha = -3^\circ, -2^\circ, -1^\circ, 0^\circ, 1^\circ, 2^\circ$

Optional:

- **Cases 3:** Tripped vs. Tully Turbulent
- **Cases 4:** M_∞ sweep, $C_L = .500$

Standard Grids Supplied, Custom Grids Encouraged

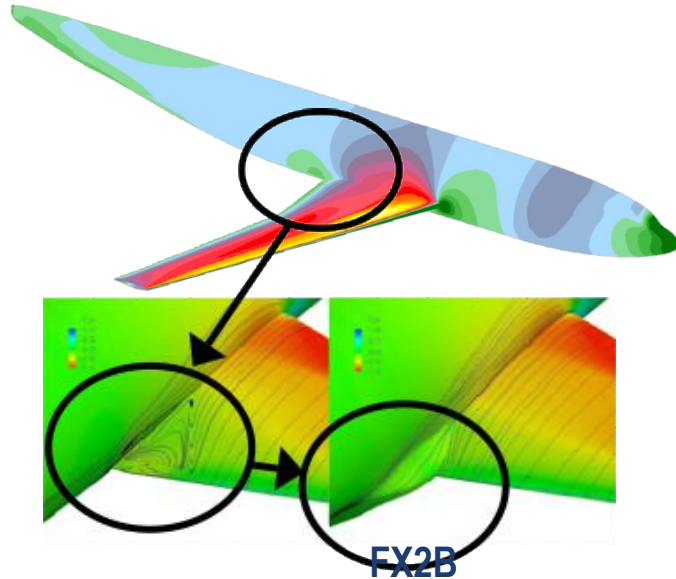
- 22 Participants
- 30 Submittals
- 20 Codes
- Grids Types
 - 14 Block Structured
 - 5 Overset
 - 11 Unstructured
- Turbulence Models
 - 16 Spalart-Allmaras (all variations)
 - 5 Menter SST
 - 3 k-omega
 - 1 Realizable k- epsilon
 - 1 k-Wilcox



- Results in general vary similar to DPW-I
- Grid convergence study envisioned, but provided grids proved insufficient (Richardson extrapolation issue)
- Compared to DPW-I, large reduction in overall spread, but dispersion of the core essentially unchanged
- Configuration Increment (pylon/nacelle) much larger than typically explored by with detailed CFD studies
- Increments more forgiving than absolutes (canceling errors)

Large side-of-body separation led to design of a fairing for DPW3 drag increment test case

DLR-6F Wind Tunnel Model



Fairing Designed to Eliminate Side-of-Body Separation

Configuration 1 - DLR-F6 Wing-Body with and without FX2B fairing

- Grid Convergence $M_\infty = 0.75$, $C_L = 0.500$
- Alpha Sweep $M_\infty = 0.75$, $\alpha = -3^\circ$ to 1.5°

Simple Wings



Same wings featured in DPW8

Configuration 2 - Wing Alone Comparison

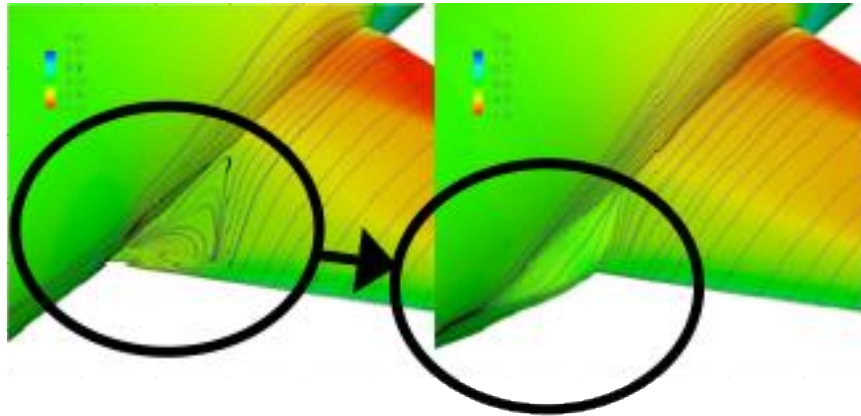
DPW-W1 - baseline

DPW-W2 - simple optimization

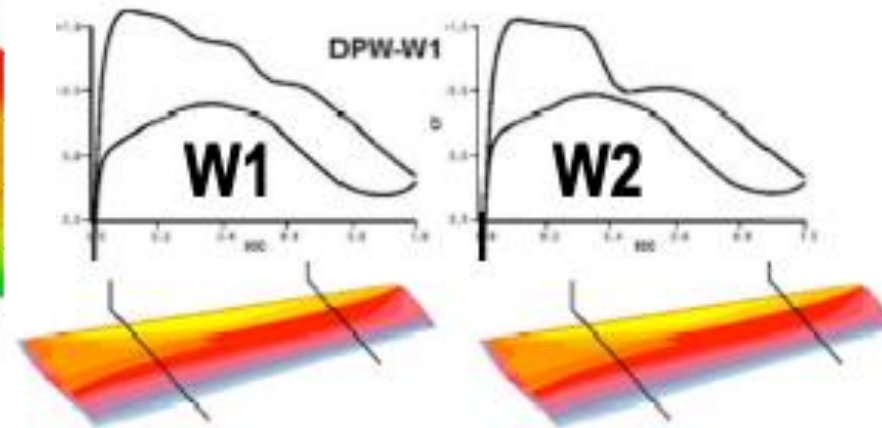
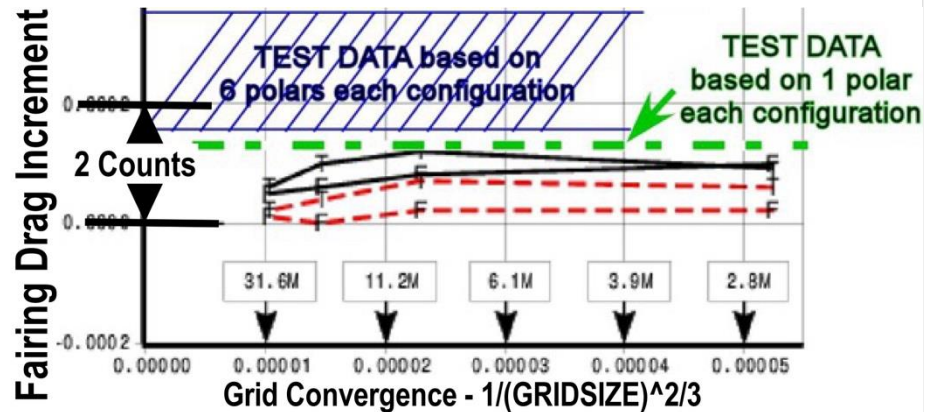
- Grid convergence study at $Mach=0.76$, $\alpha=0.5$
- Drag polar at $Mach=0.76$, $\alpha = -1$ to 3°

Standard Grids Supplied, Custom Grids Encouraged

- | | |
|-----------------------|-----------------------|
| • 15 Participants | • Turbulence Models |
| • 30 Submittals | – 12 Spalart-Allmaras |
| • 18 Codes | • (all variations) |
| • Grids Types | – 8 Menter SST |
| – 11 Block Structured | – 3 EASM |
| – 4 Overset | – 2 k-omega |
| – 11 Unstructured | – 2 k-epsilon |



FX2B Fairing Designed to Eliminate Side-of-Body Separation



- Inboard separation bubble continues to be a major source of difficulty
- No major trends based on grid type or turbulence model
- Grid convergence characteristics suggest difficulty in generating consistent sets of grids
- Some sets had obvious problems with convergence, matching CL

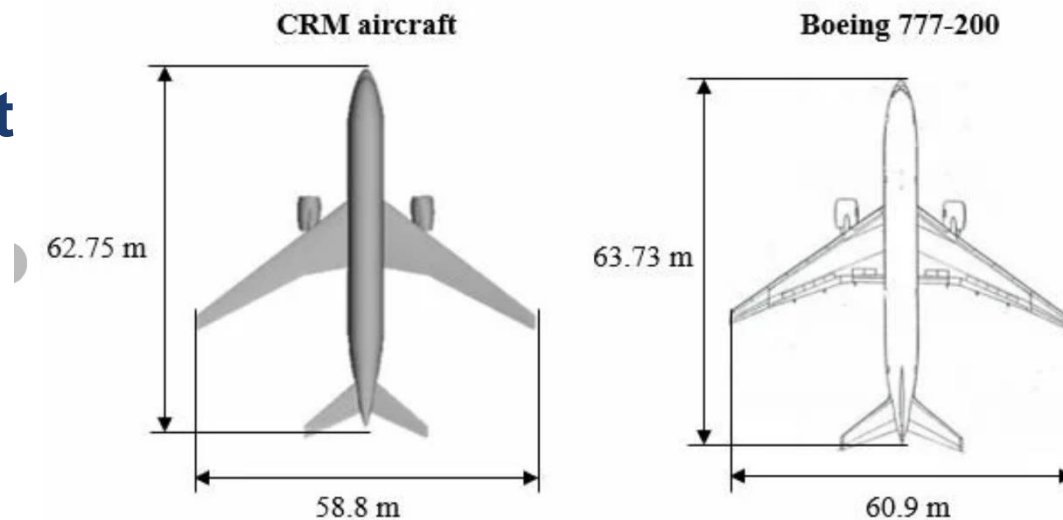
Good news

- Just about all solution sets showed the fairing to reduce drag

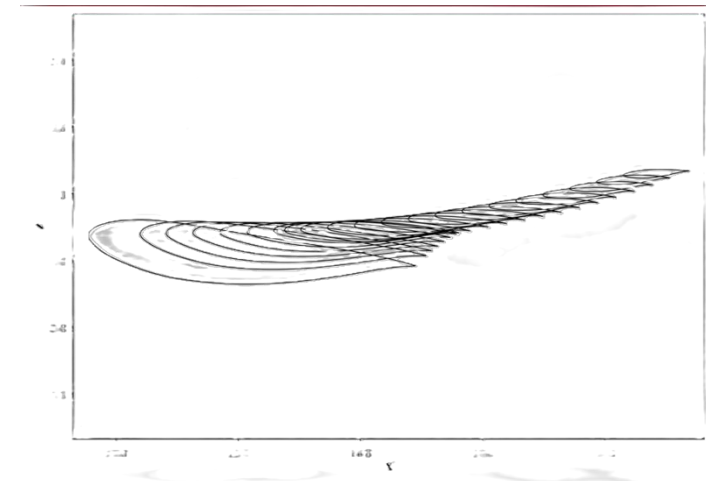
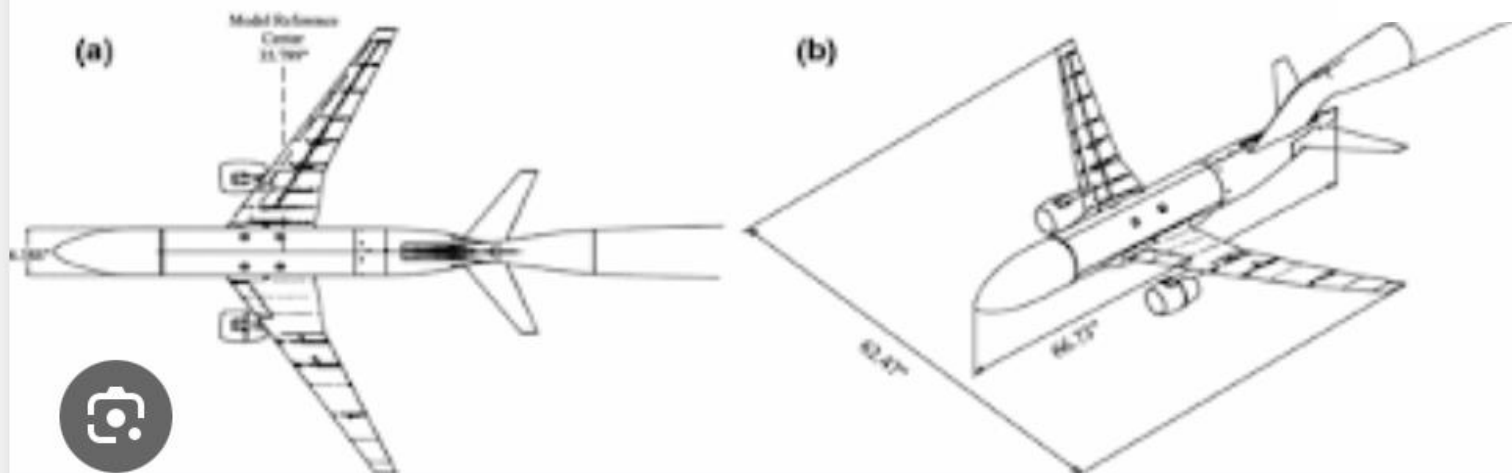
NASA High Speed Common Research Model (CRM)

- Developed in response to a need for a generic but relevant high speed transport geometry that could be used by anyone without any restrictions.
- Geometry and wind tunnel test data are available at

<https://commonresearchmodel.larc.nasa.gov>



CRM aircraft geometry compared with a traditional Part 25 transport category aircraft (Boeing 777-200) [2].

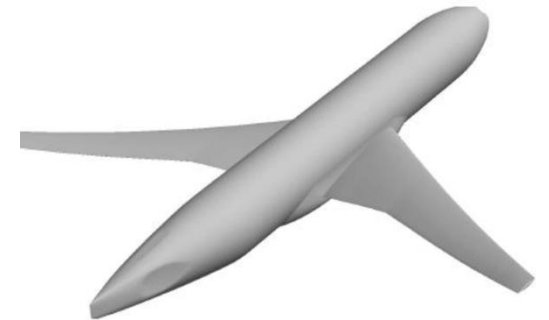


- Built to be tested in cryogenic wind tunnels (NTF & ETW)
- Different scale wind tunnel models have also been built and tested by ONERA and JAXA



Subject in

**DPW4, DPW5, DPW6,
DPW7, DPW8**



DPW4



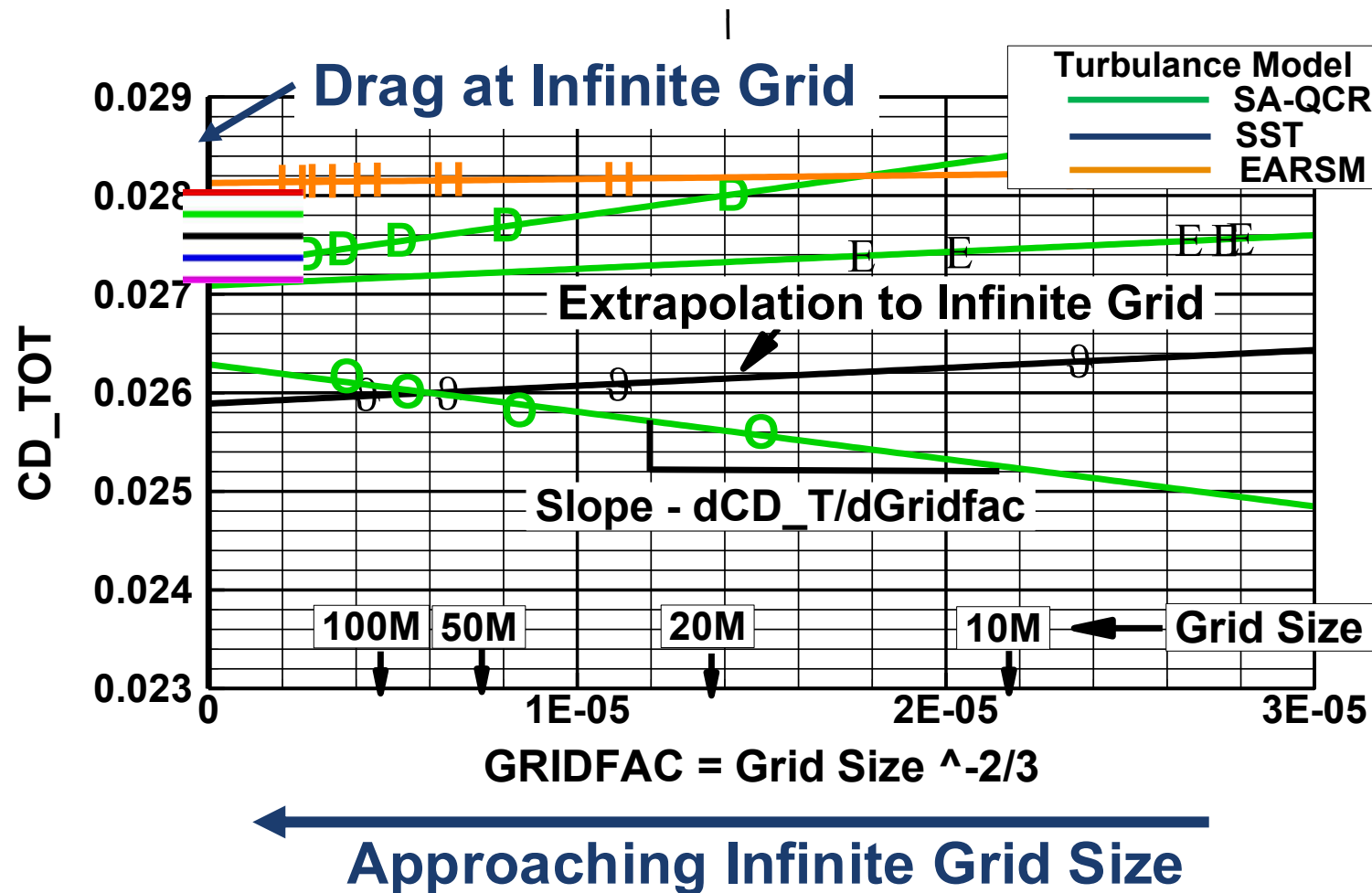
DPW6



For more detail on previous CRM DPWs: Tinoco, NASA CR Summary Report (CR-2019-220284)

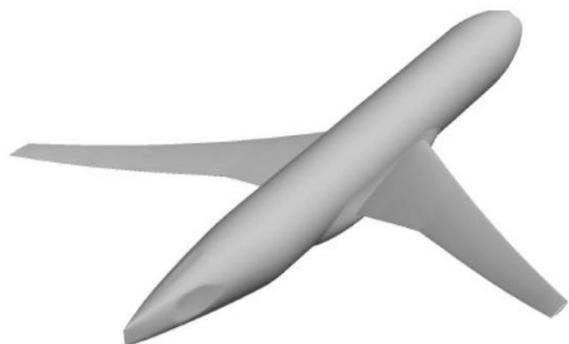
Richardson Extrapolation

- Solutions are obtained on a parametric family of three-dimensional meshes that have been “uniformly” refined in all three coordinate directions.
- Plot function vs $1/(\text{grid size}^{2/3})$ for family of grid sizes
- For 2nd order codes, should be linear vs. $\text{Grid_Factor} = N^{-2/3}$
- 0-intercept estimates theoretical infinite resolution (continuum) result

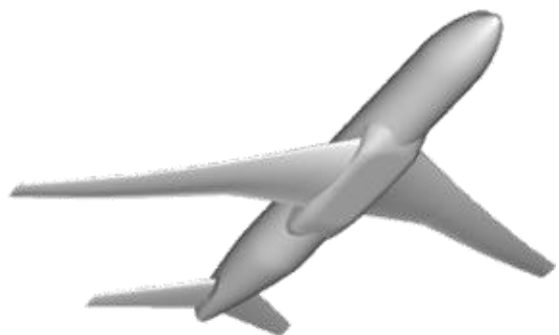


NASA High Speed Common Research Model (CRM)

CRM WingBody



CRM WingBodyTail



- **Case 1a:** Grid Convergence $M_\infty = 0.85$, $C_L = 0.500 \pm 0.005$ Tail = 0°
- **Case 1b:** Downwash Study $M_\infty = 0.85$, $\alpha = 0^\circ$ to 4° , Tail = -2° , 0° , 2° , off

Optional:

- **Cases 2:** M_∞ sweep, 0.70 to 0.87. $C_L = .400, .450, .500$ Tail = 0°
- **Cases 3:** Reynolds number study $M = 0.85$. $Re = 5M$ & $20M$

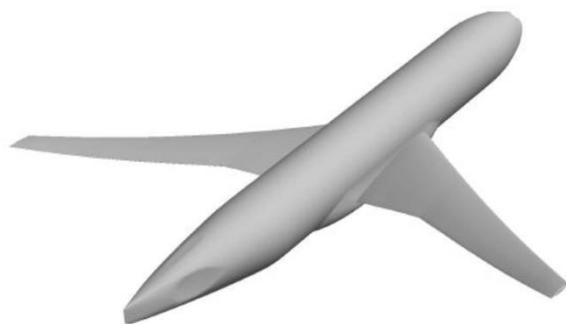
Standard Grids Supplied, Custom Grids Encouraged

18 Participants, 28 Data Submittals from 17 organizations

- Grid types: 17 Unstructured, 11 Structured
- Turbulence Models
 - 18 Spalart-Allmaras (all variations)
 - 7 Menter SST k- ω
 - 1 EARSM (Explicit Algebraic Reynolds Stress Model)
 - 1 SSG/LLR- ω (Reynolds Stress Model)
 - 1 Realizable k- ϵ

NASA High Speed Common Research Model (CRM)

CRM WingBody



- **Case 1: Grid Convergence** $M_\infty = 0.85$, $C_L = 0.500 \pm 0.001$, $Re = 5M$
- **Case 2: Angle-of-Attack Sweep** $M_\infty = 0.85$, $\alpha = 2.5^\circ$ to 4° , $Re = 5M$
- **Cases 3: Turbulence Model Verification**
 - 2D Zero Pressure Gradient Flat Plate: $M = 0.20$; $Re_L = 5 \times 10^6$; $T_{ref} = 540^\circ R$
 - 2D Bump-in-channel: $M = 0.20$; $Re_L = 3 \times 10^6$; $T_{ref} = 540^\circ R$
 - 2D NACA 0012 Airfoil: $M = 0.15$; $Re_C = 6 \times 10^6$; $T_{ref} = 540^\circ R$

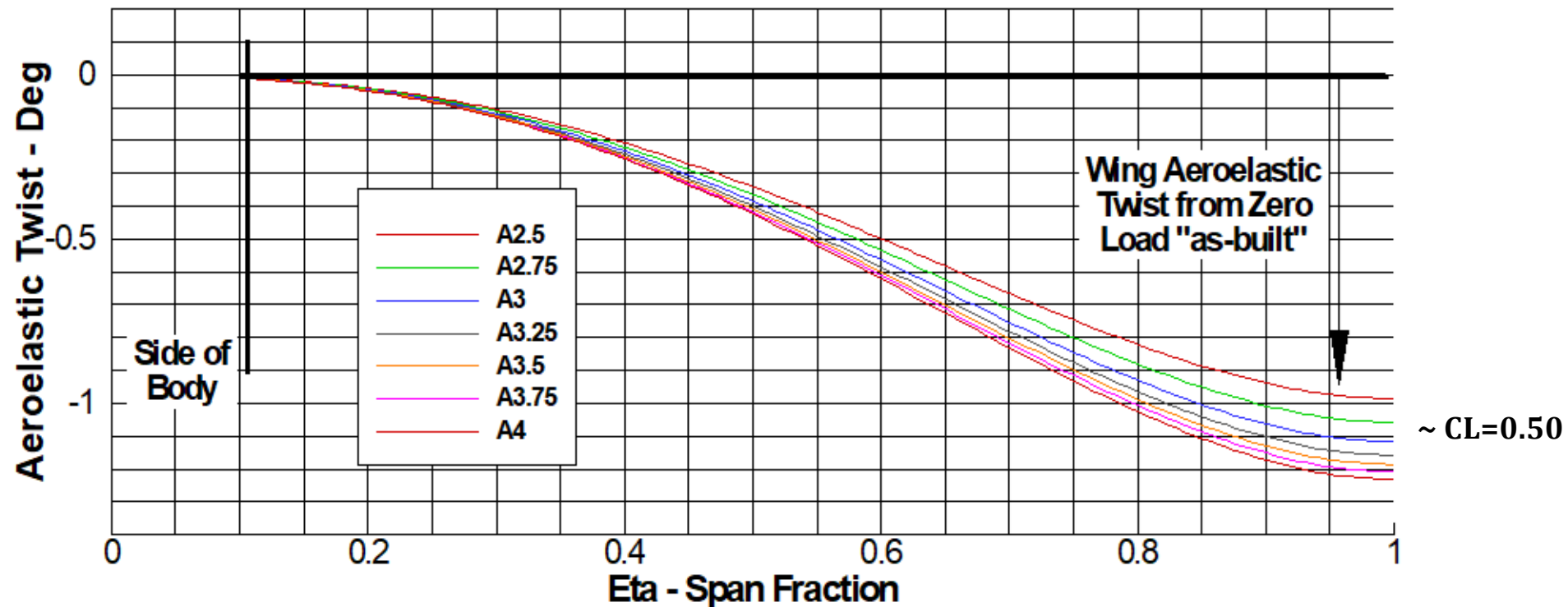
Standard Grids Supplied, Custom Grids Encouraged

27 Participants, 57 Data Submittals from 22 organizations

- **Grid types: 25 Unstructured, 12 Structured**
- **Turbulence Models**
 - 38 Spalart-Allmaras (all variations)
 - 13 Menter SST k-omega
 - 1 EARSM (Explicit Algebraic Reynolds Stress Model)
 - 1 Lag-RSTSA
 - 4 Goldberg R_T

Static Aeroelastic Wing Twist Distribution Derived from Wind Tunnel Measurements

Mach = 0.85, Dynamic Pressure = 1385 psf

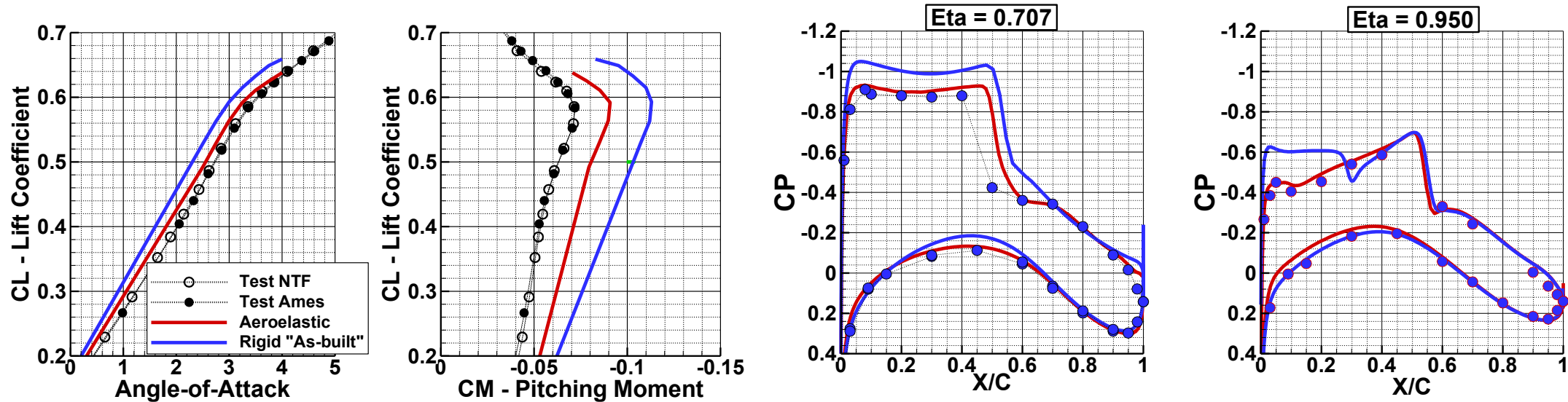


Model geometry and grids defined for each angle of attack

Importance of Aeroelastics at Transonic Speeds

- Wind Tunnel model aeroelastic deformation is significant

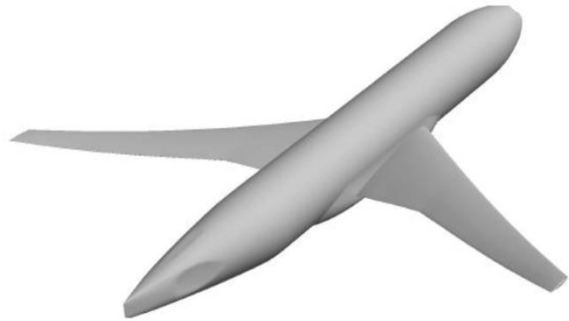
Static Aeroelastic (based on WT test measurements)
Rigid "As-Built"



- Available geometries and grids include:
 - Wing measured aeroelastic twist and deflection
 - Range of angles-of-attack and dynamic pressures
- <https://dpw.larc.nasa.gov>

NASA High Speed Common Research Model (CRM)

CRM WingBody



- Case 1: Turbulence Model Verification

2D NACA 0012 Airfoil: $M = 0.15$; $Re_C = 6 \times 10^6$; $T_{ref} = 540^\circ R$

- Case 2: Grid Convergence / Nacelle-Pylon Drag Increment

$M_\infty = 0.85$, $C_L = 0.500 \pm 0.001$, $Re = 5M$

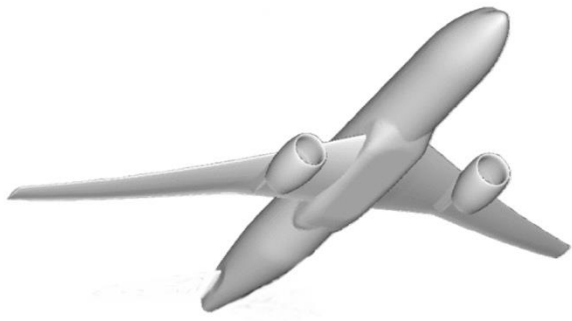
- Case 3: Angle-of-Attack Sweep $M_\infty = 0.85$, $\alpha = 2.5^\circ$ to 4° , $Re = 5M$

- Case 4: CRM WB Grid Adaptation [Optional]

- Case 5: CRM WB Coupled Aero-Structural Simulation [Optional]

Standard Grids Supplied, Custom Grids Encouraged

CRM WingBody PylonNacelle



27 Participants, 55 Data Submittals from 25 organizations

- Grid types: 26 Unstructured, 11 Structured, 7 Cartesian

- Turbulence Models

- 28 Spalart-Allmaras (all variations)

- 3 Menter SST k-omega

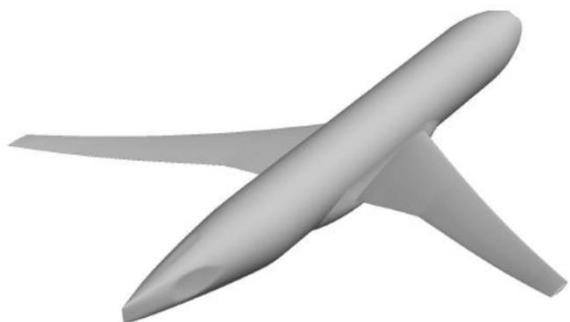
- 1 EARSM (Explicit Algebraic Reynolds Stress Model)

- 1 RSM-w

- 5 k-e

NASA High Speed Common Research Model (CRM)

CRM WingBody



Mach – 0.85, Re=5M, Re=20M

- Case 1: Grid Convergence Study
- Case 2: Angle-of-Attack Sweep
- Case 3: Reynolds Number Sweep

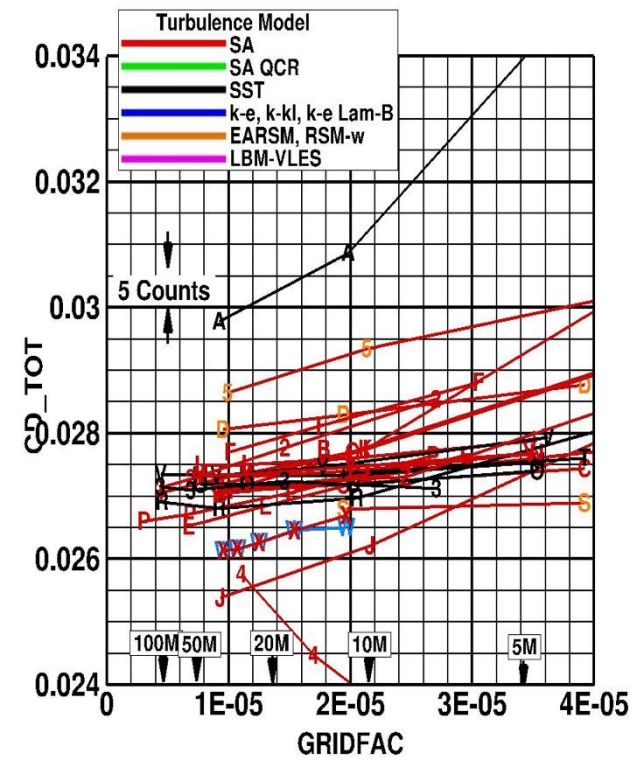
Low Q, Re=5M, Re=20M: Hi Q, Re=20M, Re=30M:
(Optional)

- Case 4: Grid Adaptation
- Case 5: Beyond RANS)- **Insufficient input**
- Case 6: Coupled Aero-Structural Simulation

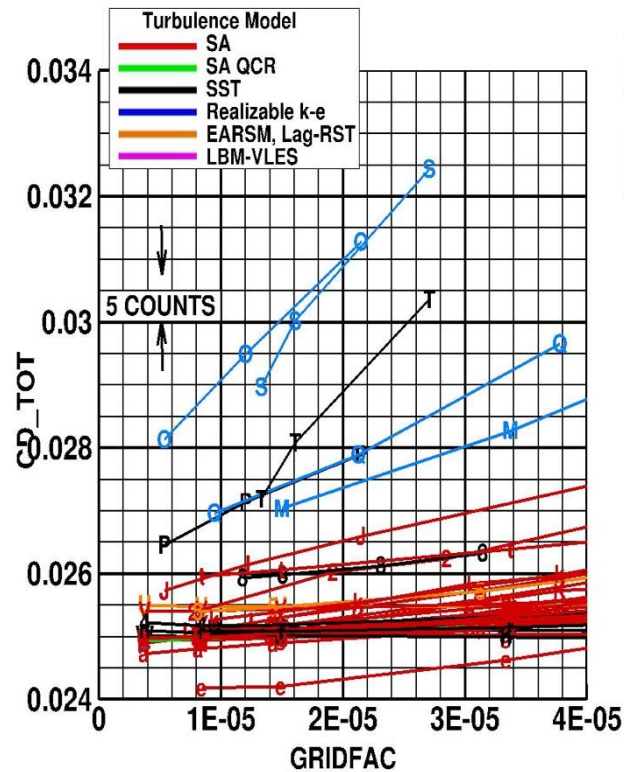
- 34 Total Data Submittals
- 18 Teams/Organizations
 - 7 N. America, 7 Europe, 4 Asia
 - 7 Government, 3 Industry, 4 Academia, 4 Commercial
- Grid Types:
 - 21 Unstructured (12 Teams)
 - 8 Structured Multi-block (5 Teams)
 - 4 Overset (3 Teams)
 - 1 Custom Cartesian (1 Team)
- Turbulence Models:
 - 16 SA-QCR (all types)
 - 7 SA w/o QCR
 - 5 SST
 - 3 EARSM
 - 1 SSG/LRR, 1 AMM-QCR, 1 RSM-In(w)

GRID Convergence – Turbulence Model

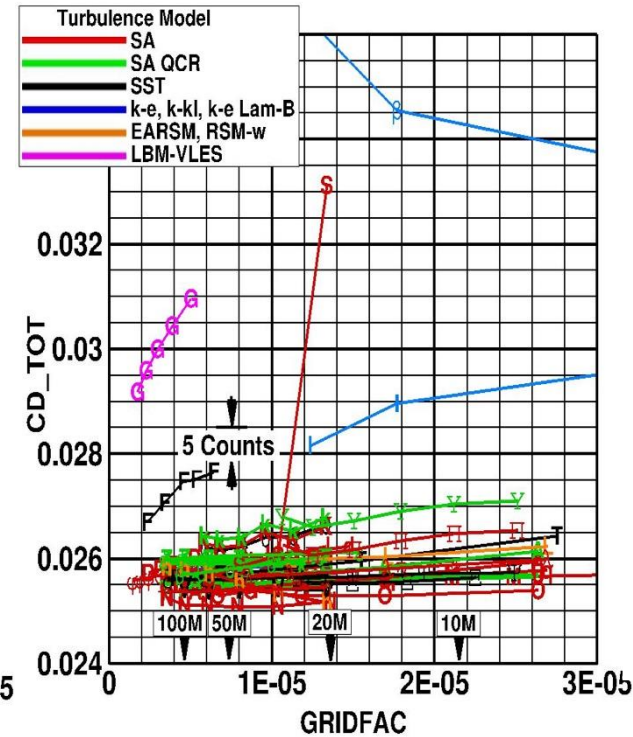
DPW4
Wing-Body-Tail
Re=5M
2009



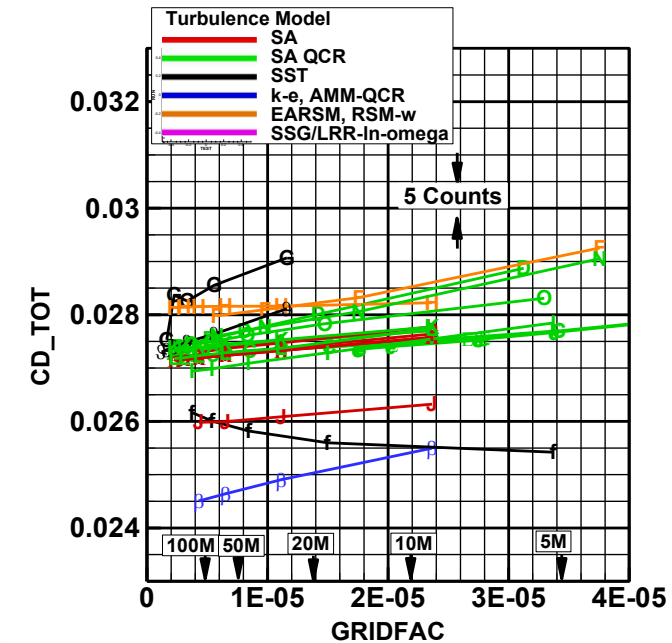
DPW5
Wing-Body
Re=5M
2012



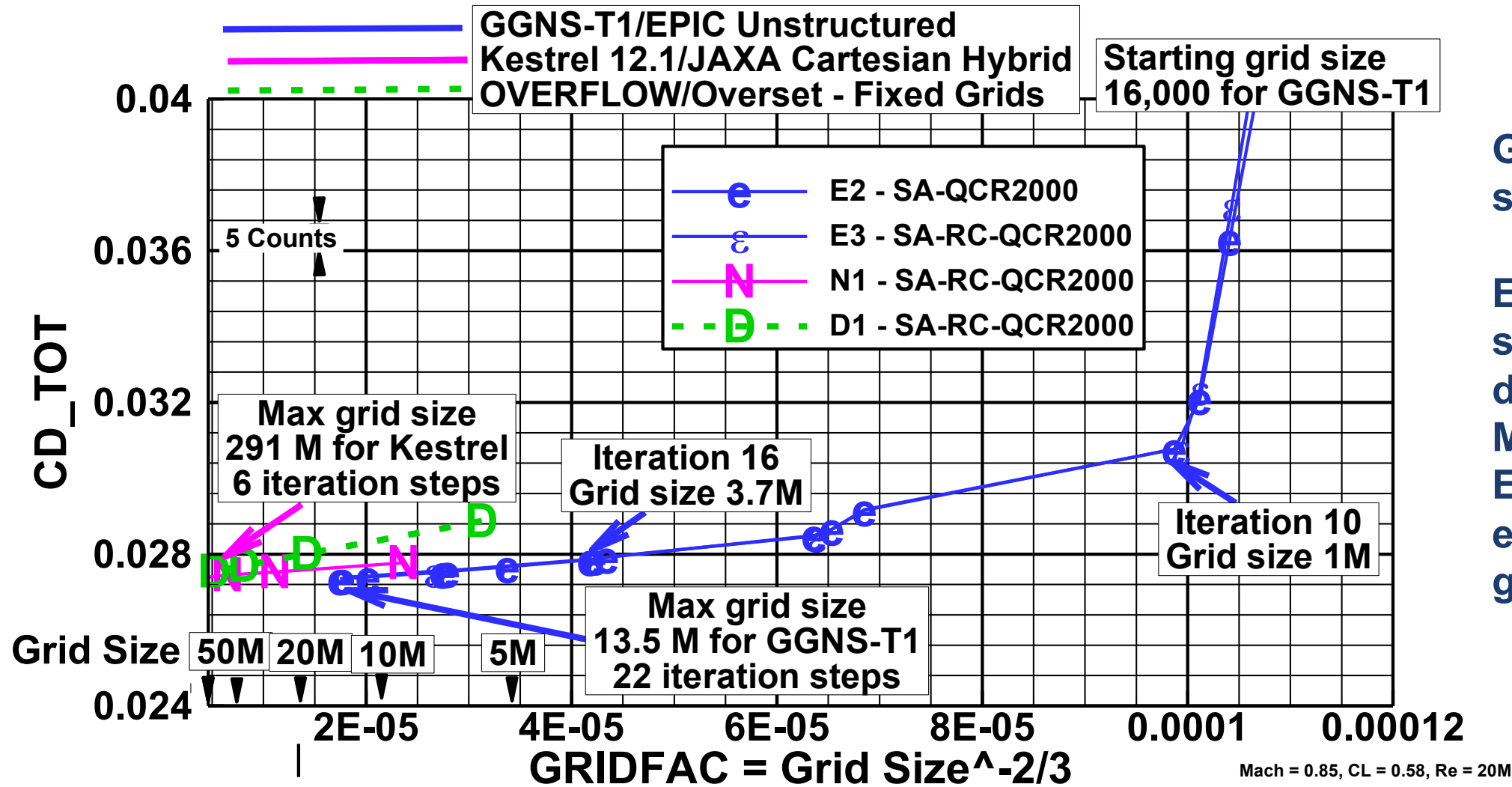
DPW6
Wing-Body
Re=5M
2016



DPW7
Wing-Body
Re=20M
2022



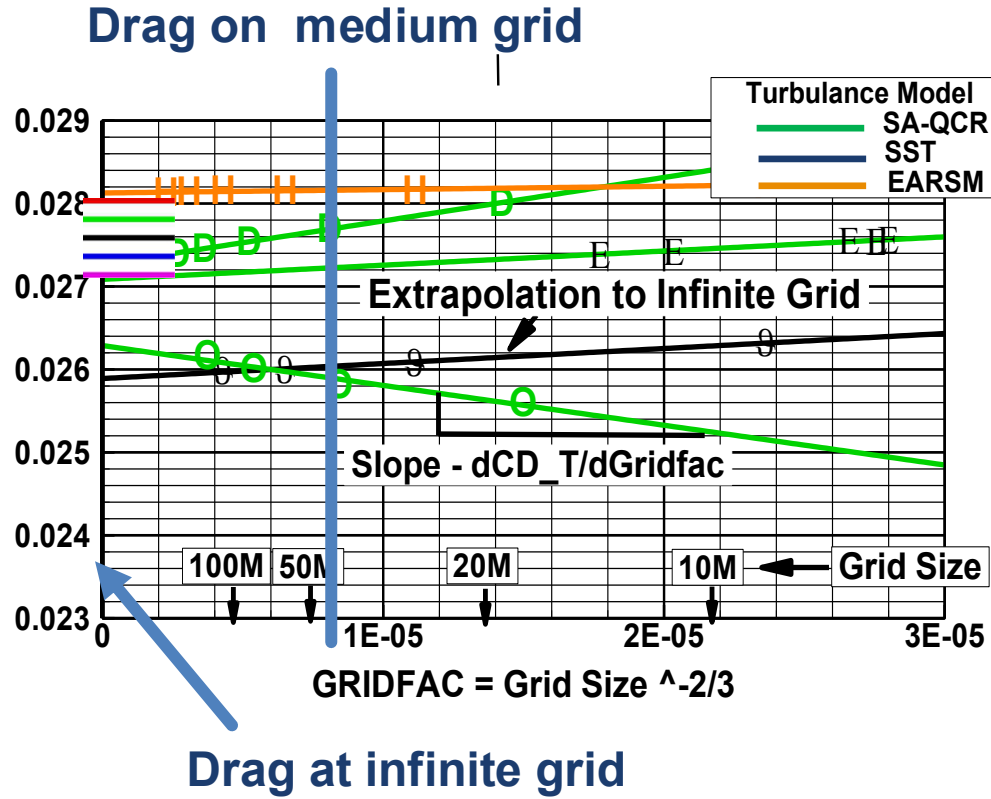
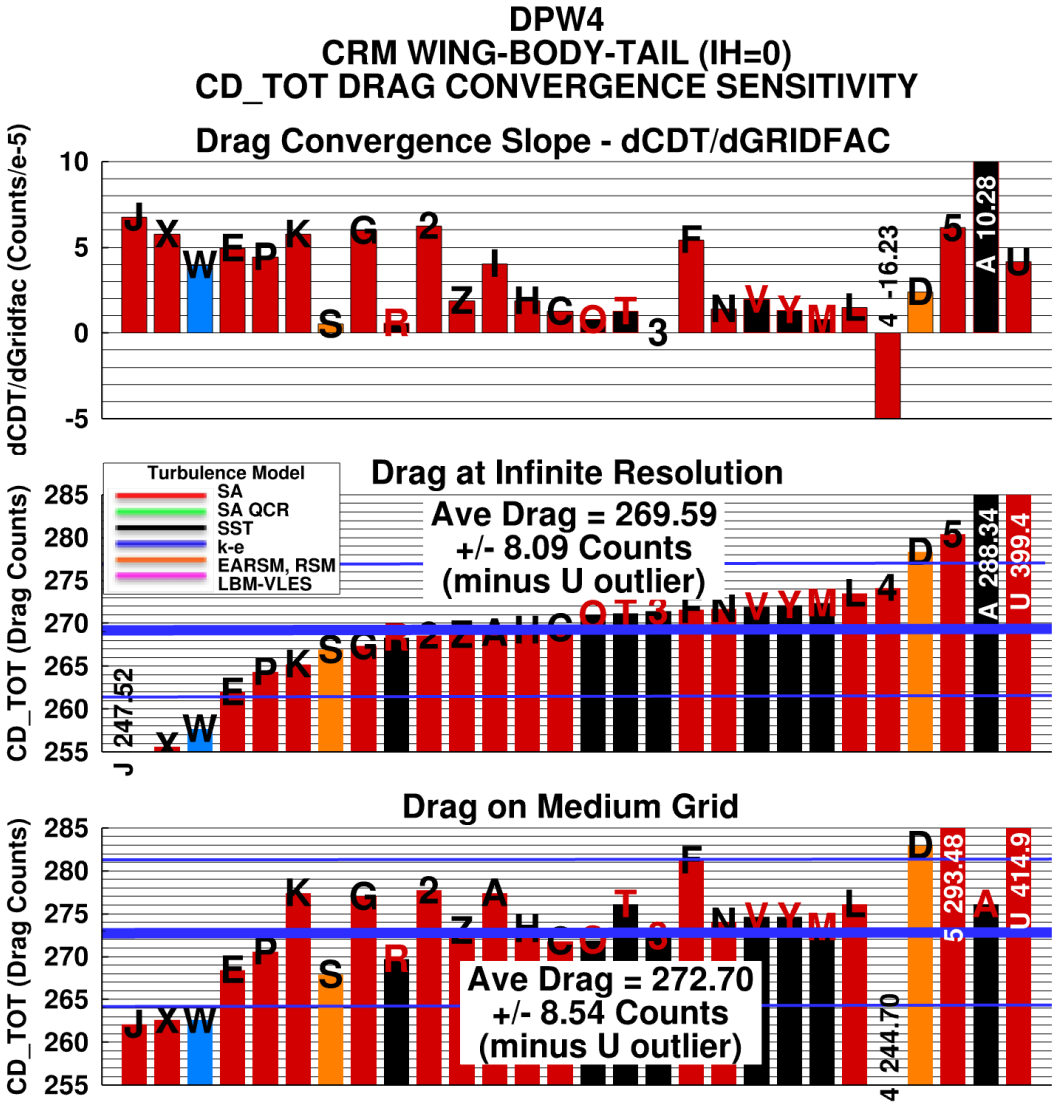
Grid Convergence - Adaptive Solutions



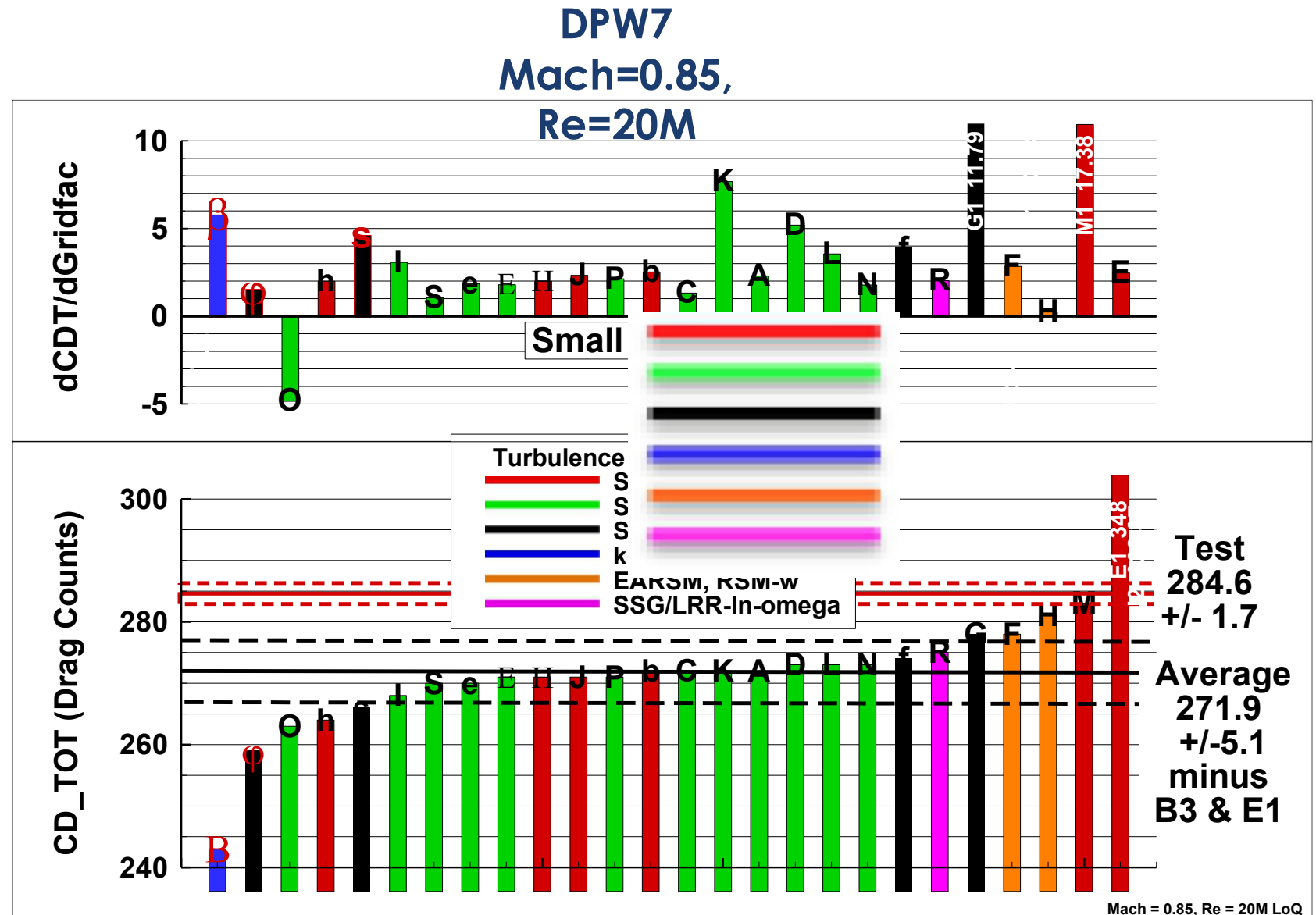
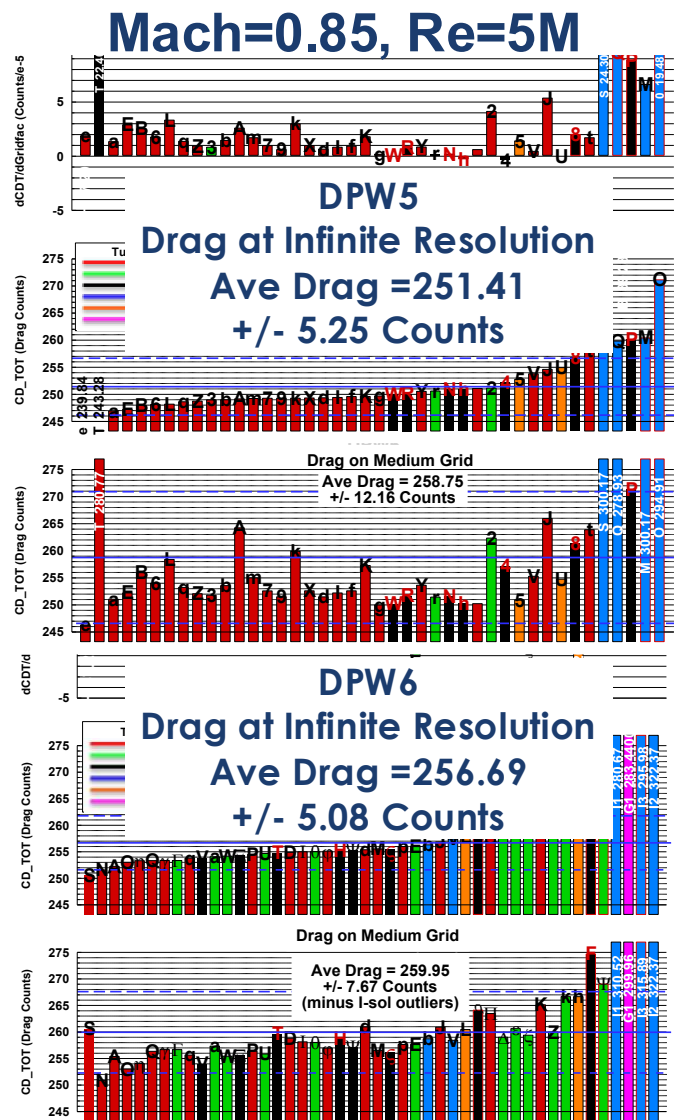
GGNS uses a strong solver

EPIC uses a sizing metric derived from the Mach Hessian or Entropy Adjoint error to drive grid adaption.

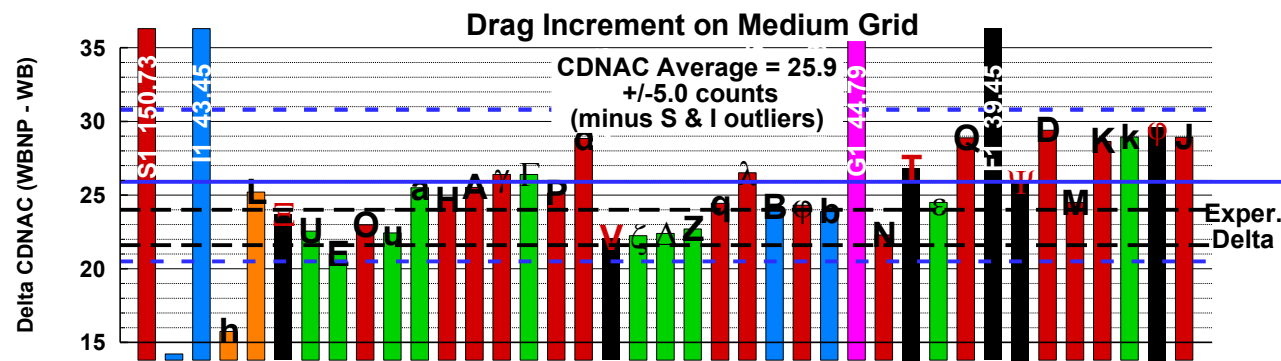
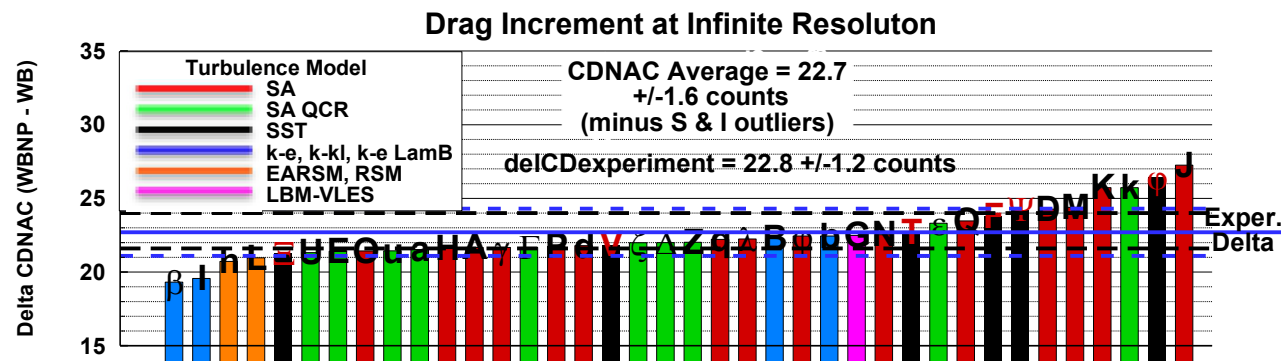
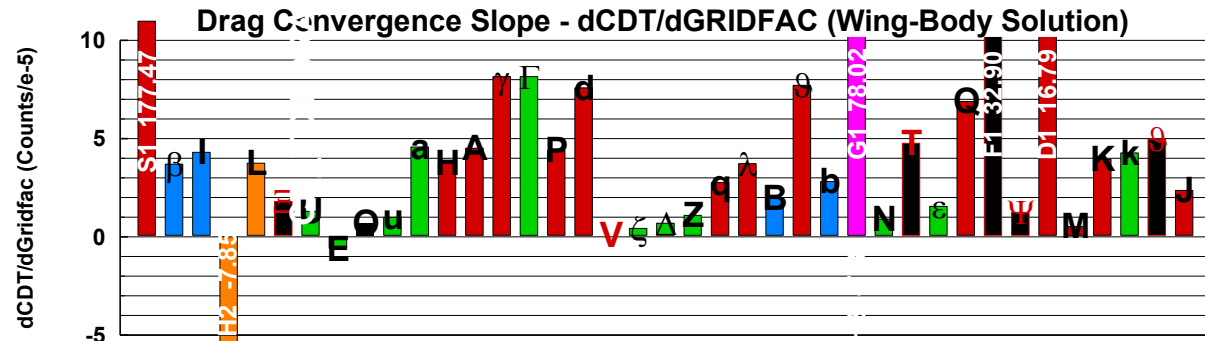
Total Drag and Drag Sensitivity Convergence



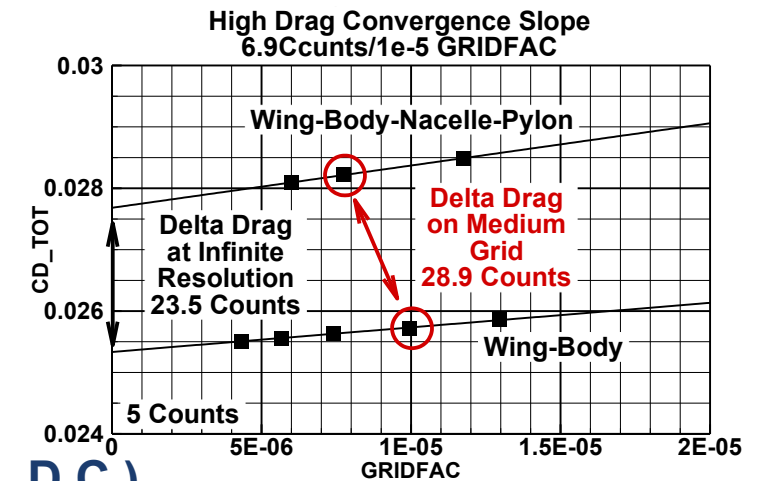
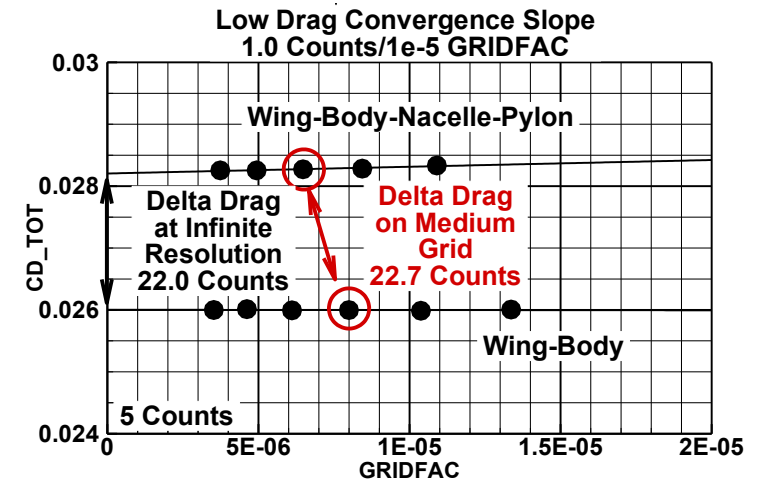
Total Drag and Drag Sensitivity Convergence



Nacelle-Pylon Drag Increment

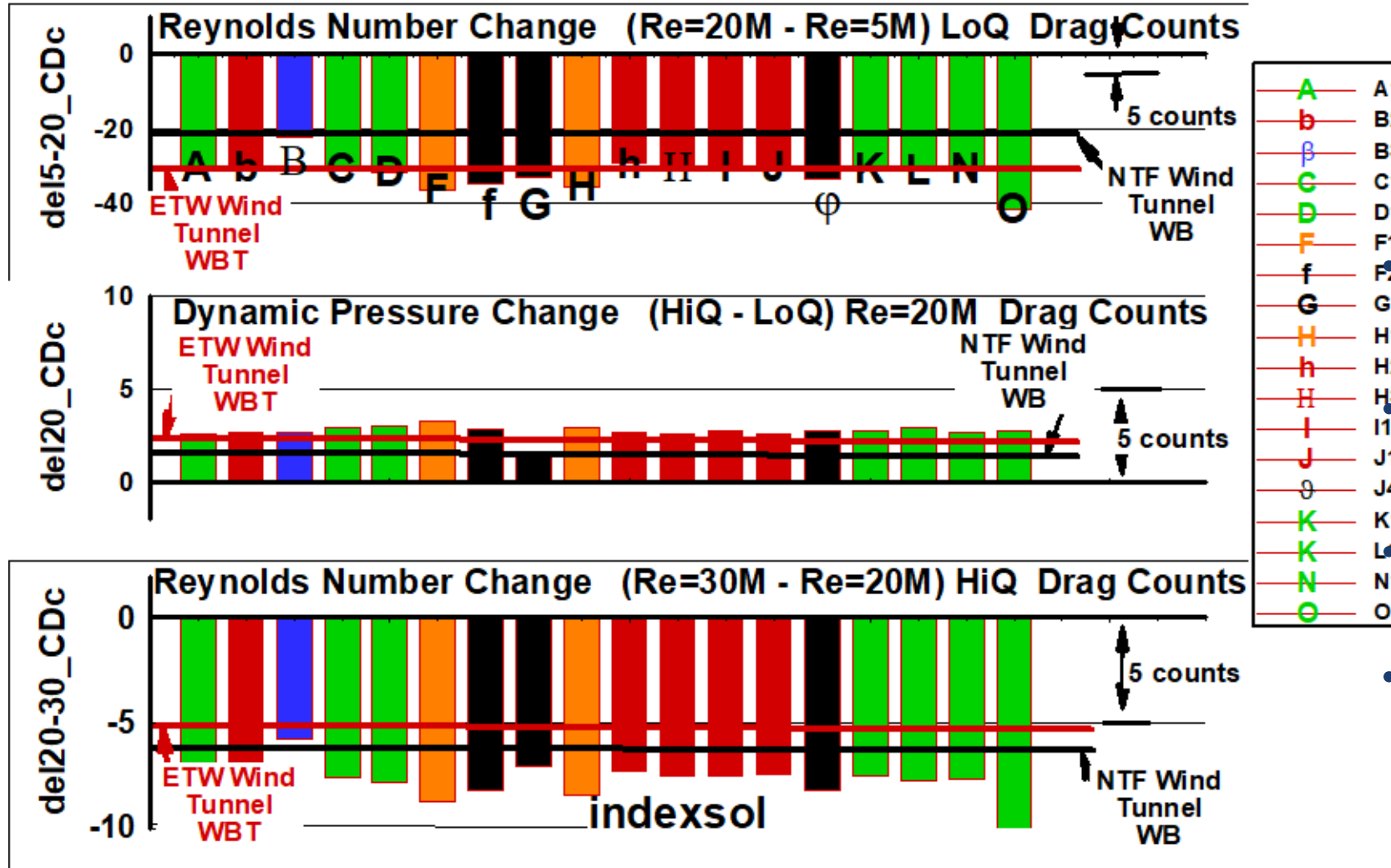


Potential Impact of Drag Convergence Slope on Drag Increment



6th CFD Drag Prediction Workshop – 2016 (Washington, D.C.)

Reynolds Number Sweep at Constant CL



Re = 5M to Re = 30M
M = 0.85, CL = 0.50 (Cruise)

Re = 5M, LoQ – R5 grid using
2.50-deg LoQ AE CRM

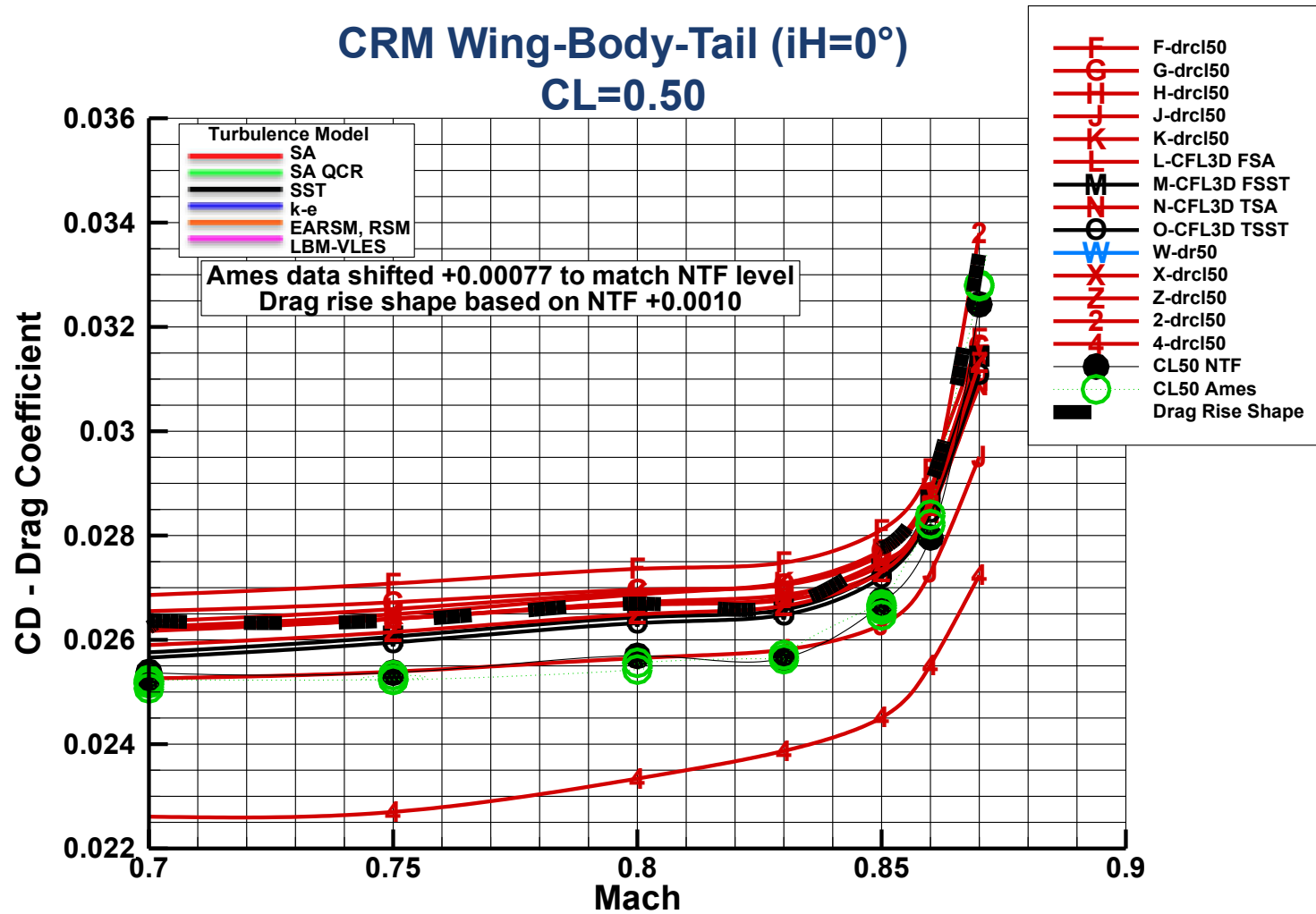
Re=20M, LoQ – R30 grid using
2.50-deg LoQ AE CRM

Re=20M, HiQ – R30 grid using
2.50-deg HiQ AE CRM

- Re=30M, HiQ – R30 grid using
2.50-deg HiQ AE CRM

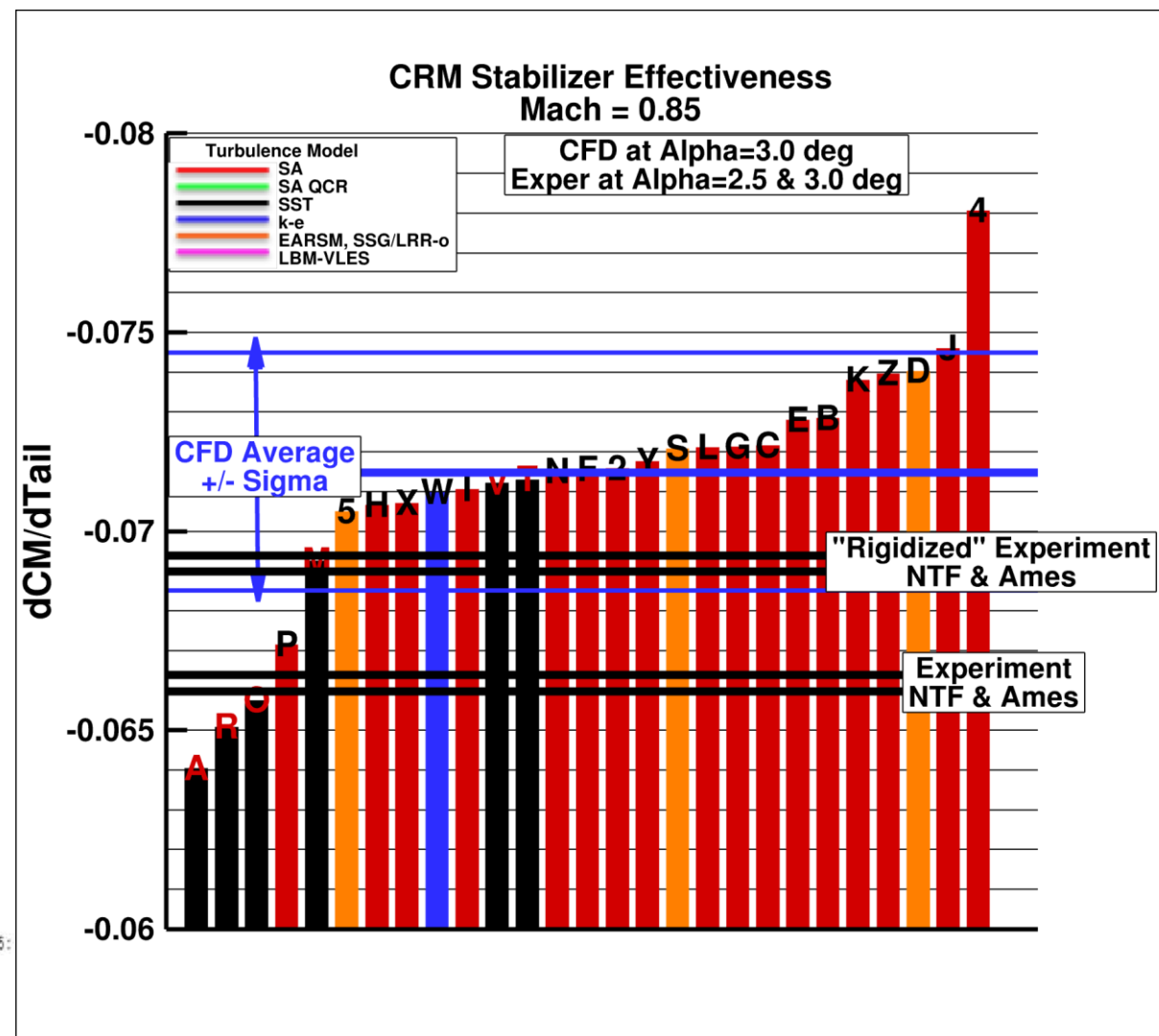
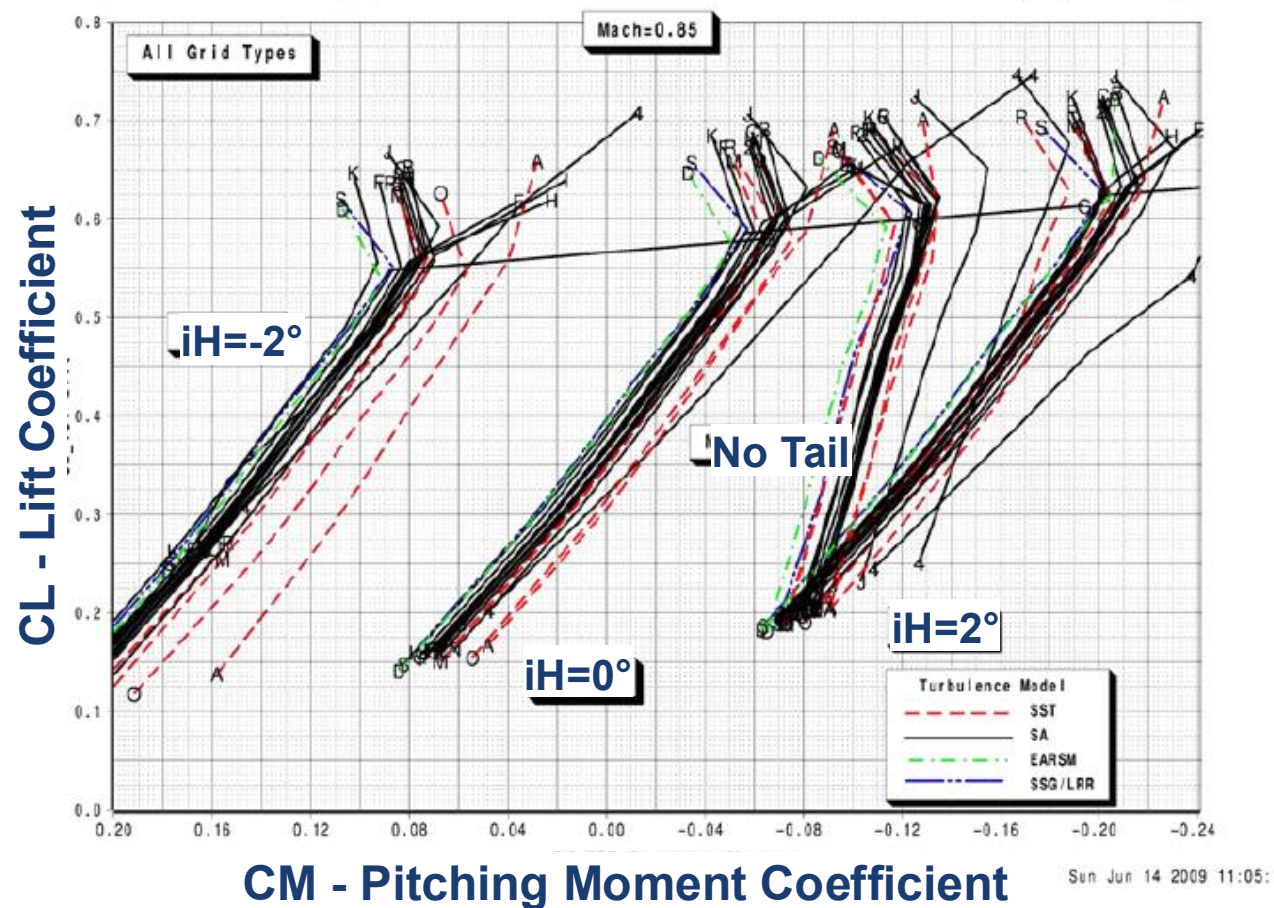
7th CFD Drag Prediction Workshop – 2022(Chicago)

Drag Rise



4th CFD Drag Prediction Workshop – 2009 (San Antonio)

Mach=0.85, Re=5M



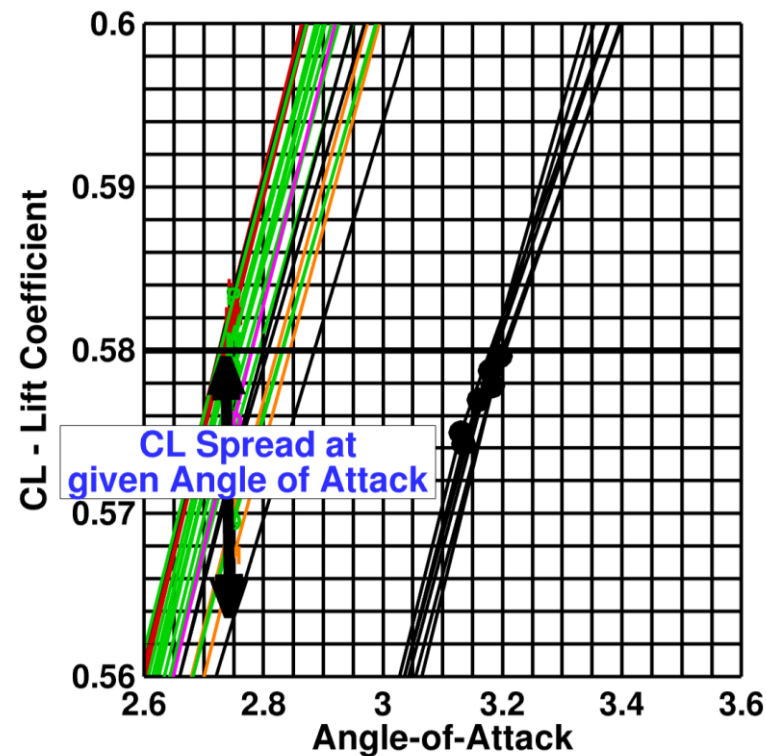
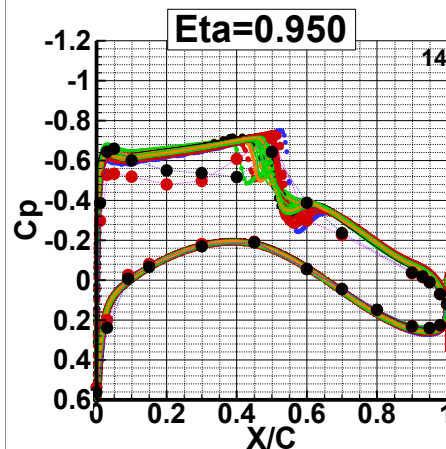
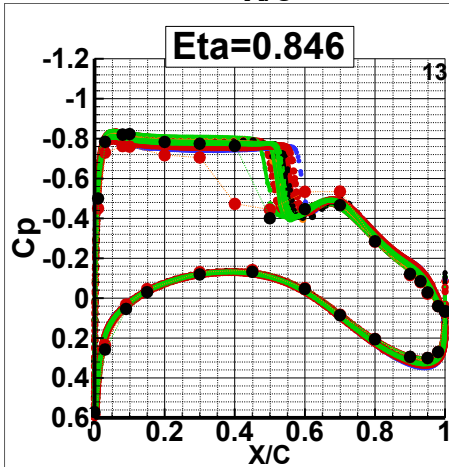
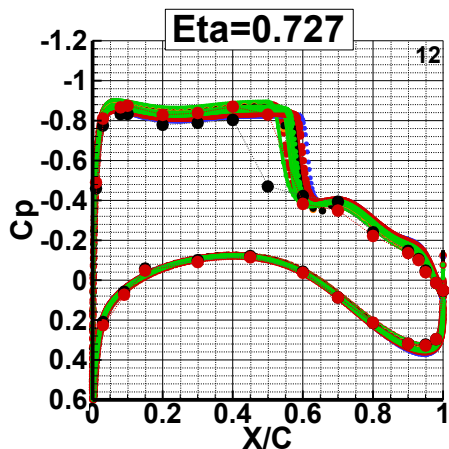
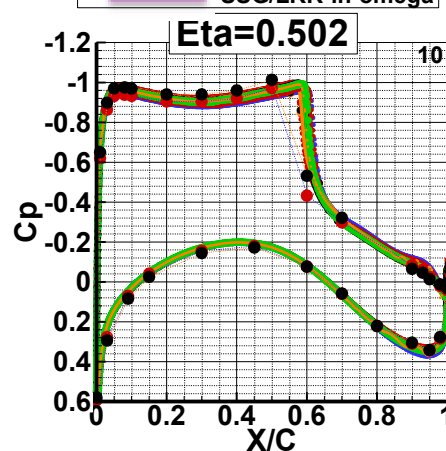
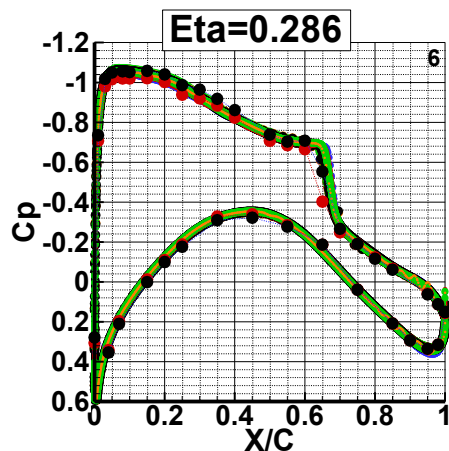
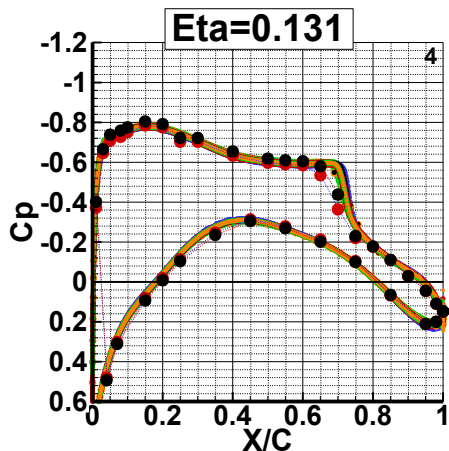
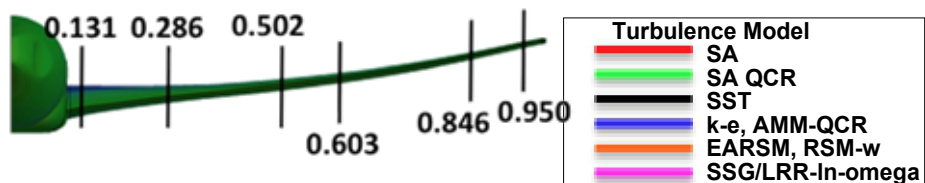
4th CFD Drag Prediction Workshop – 2009 (San Antonio)

Wing Pressure Distributions – All Solutions

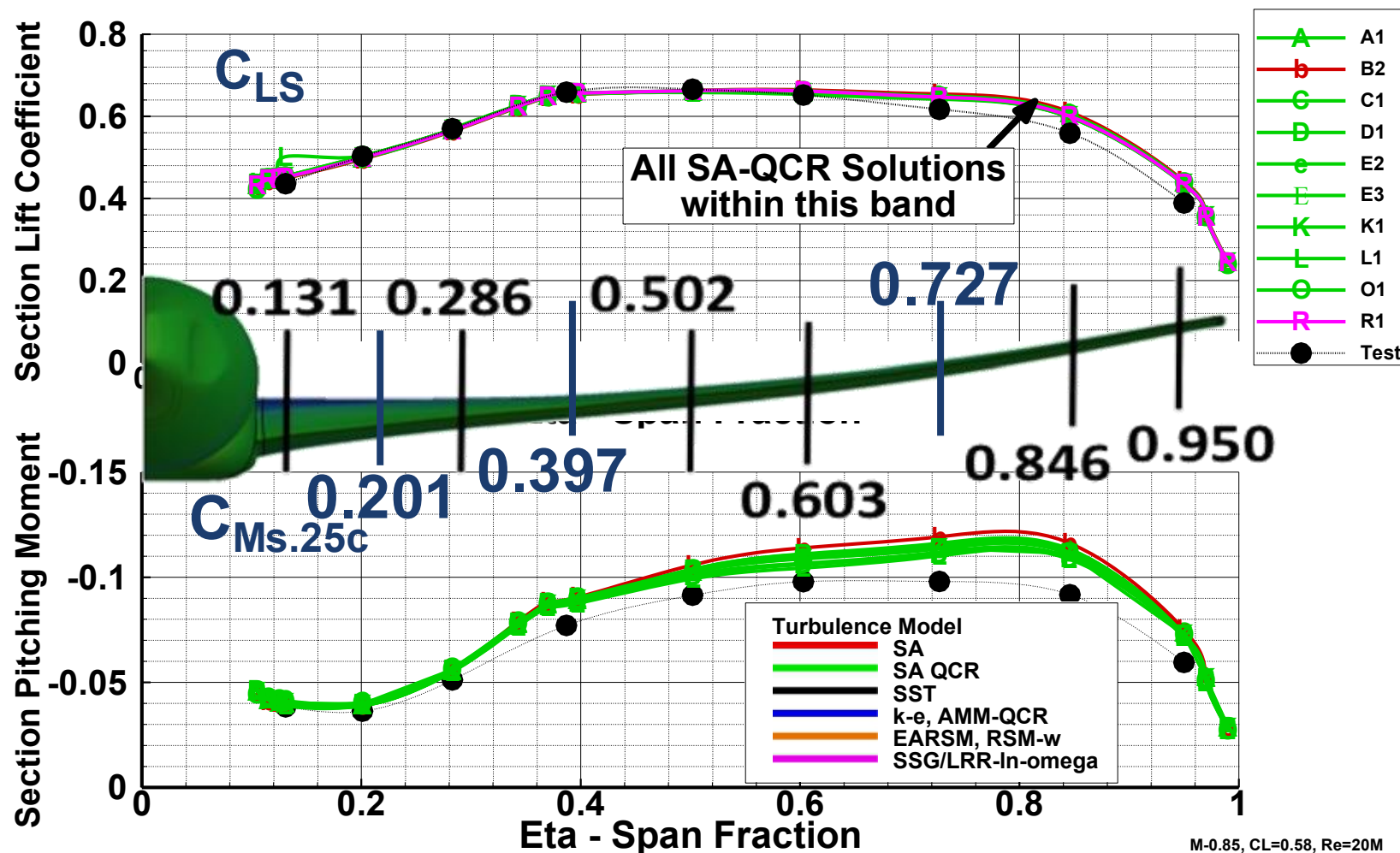
$M=0.85$, $CL=0.58$, $Re=20M$

● Test - $AOA=2.92^\circ$, $CL=0.544$

● Test - $AOA=3.16^\circ$, $CL=0.577$



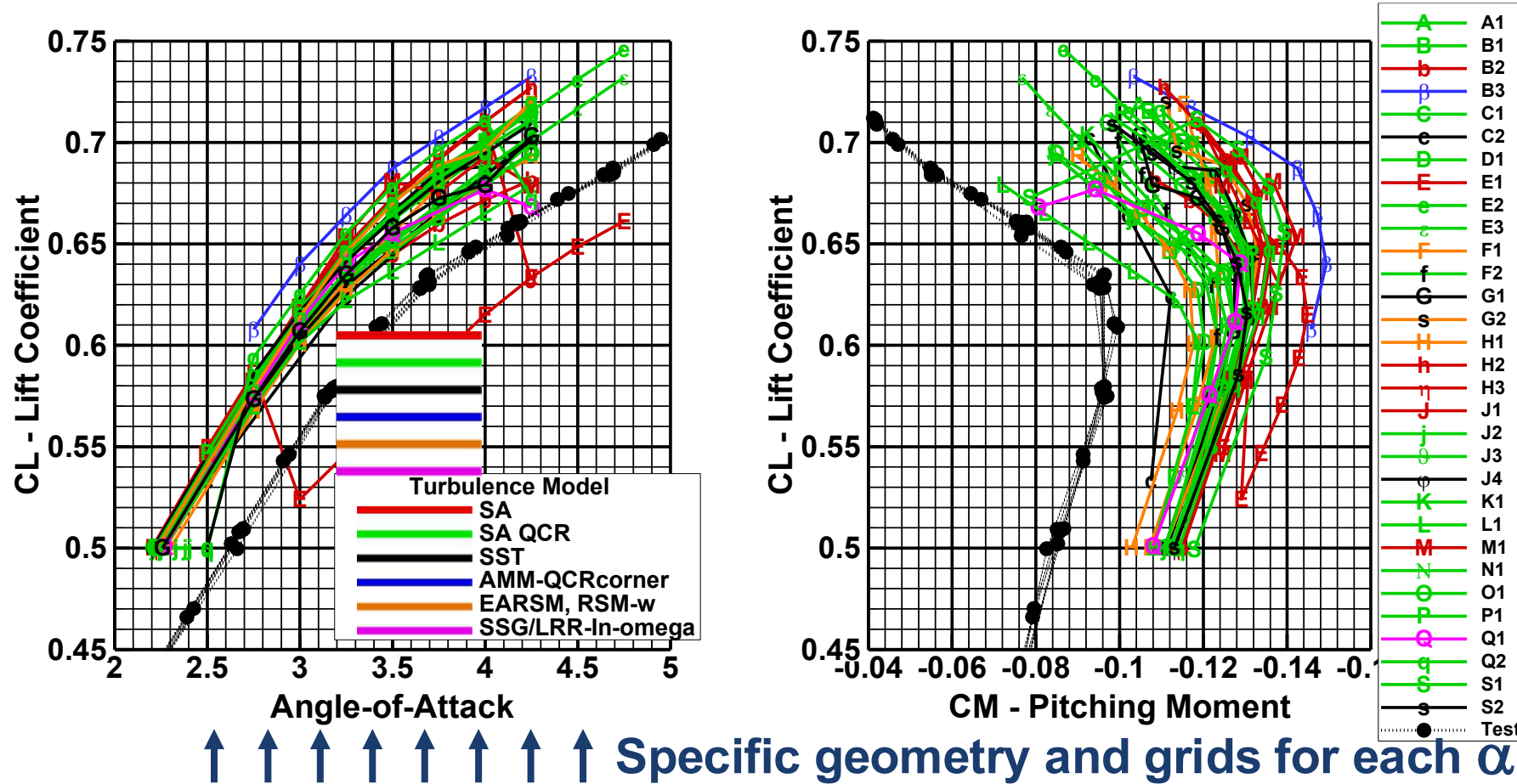
Mach = 0.85, CL = 0.58, Re = 20M



7th CFD Drag Prediction Workshop – 2022(Chicago)

Lift and pitching moment for all solutions

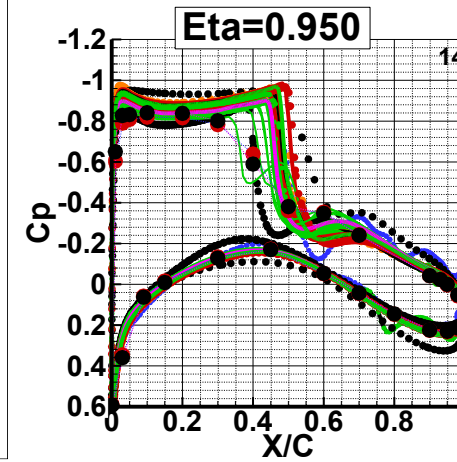
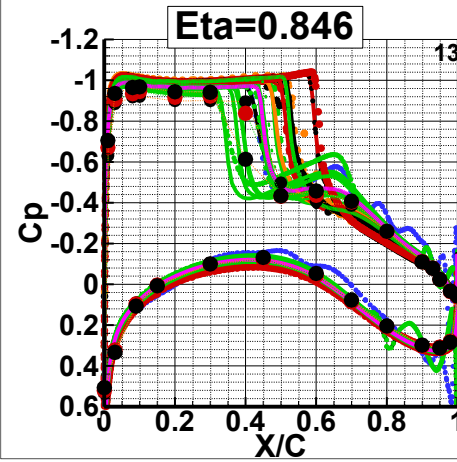
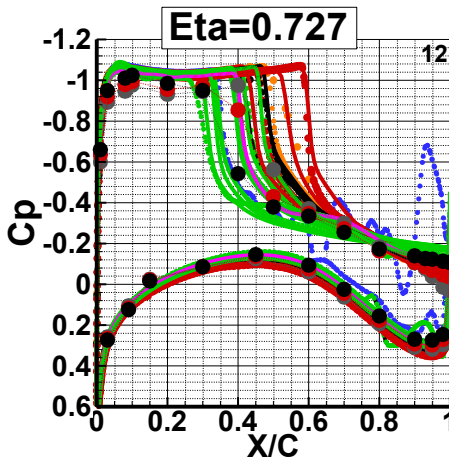
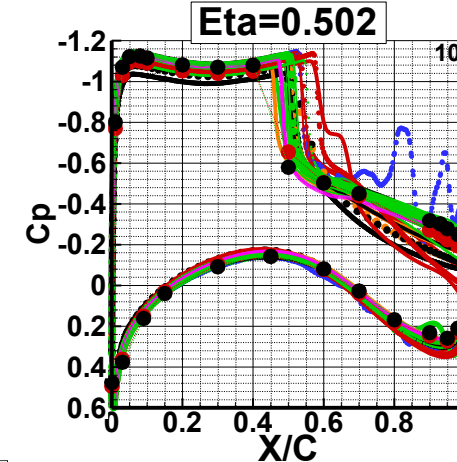
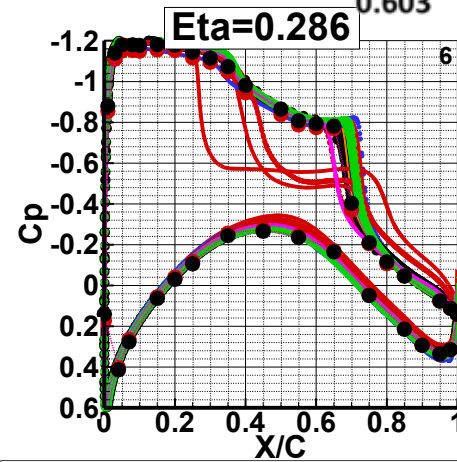
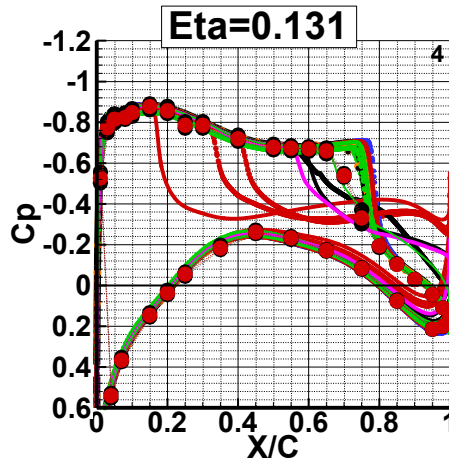
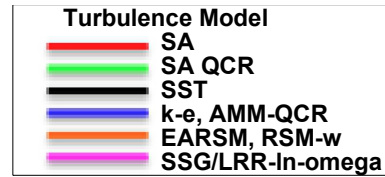
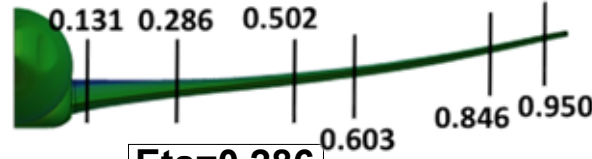
Mach = 0.85, Re = 20M

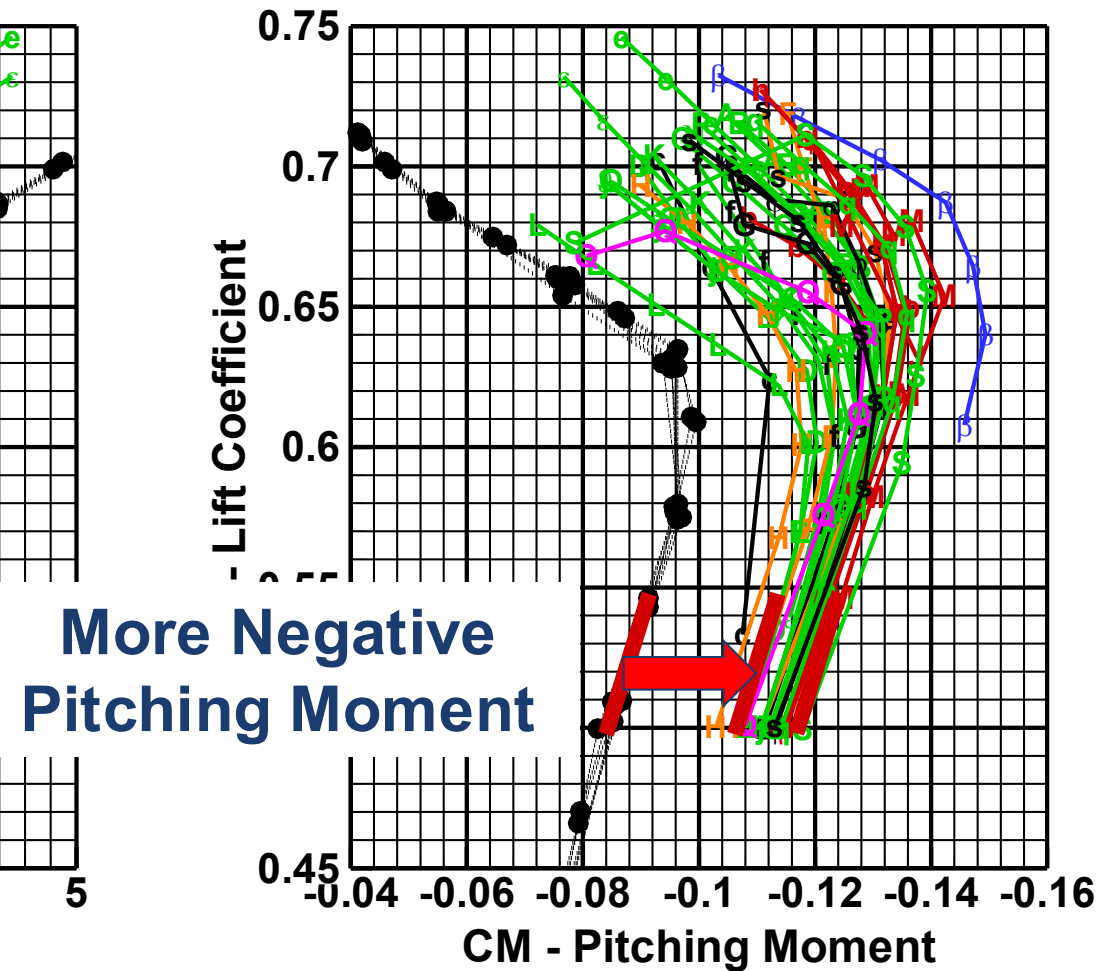
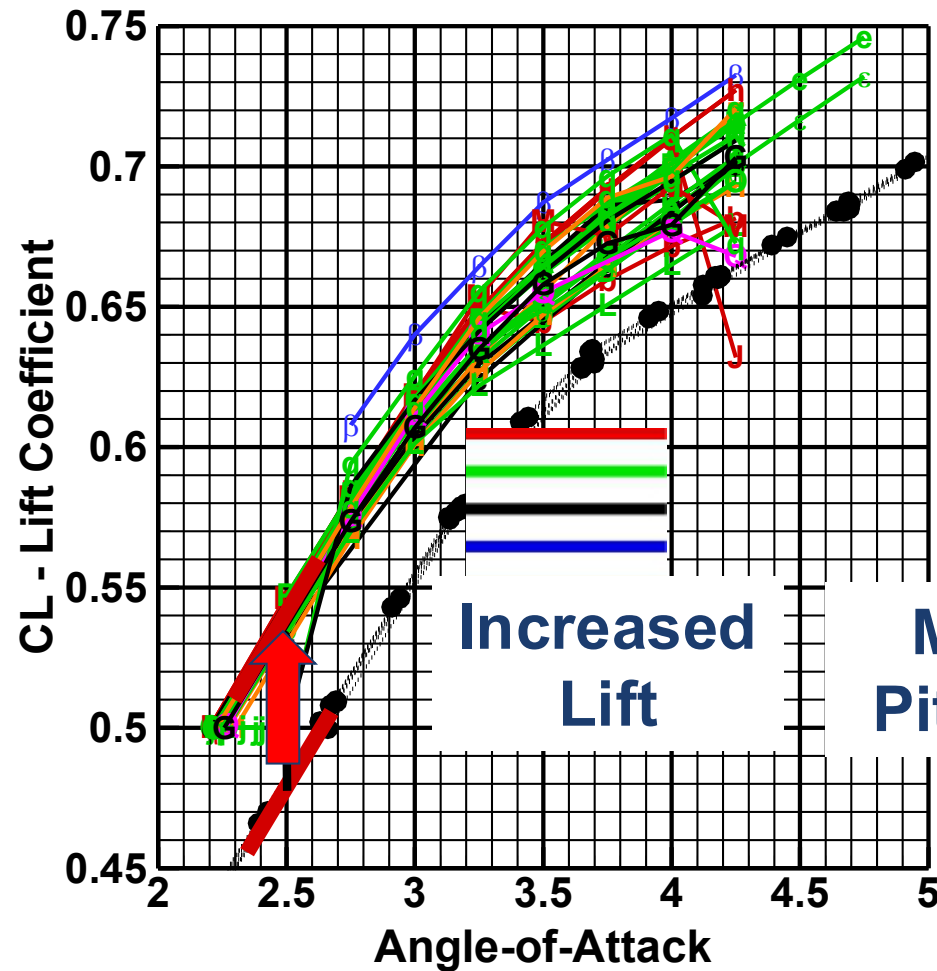


Wing Pressure Distributions – 4.25° angle-of-attack

M=0.85, Angle-of-Attack=4.25°, Re=20M

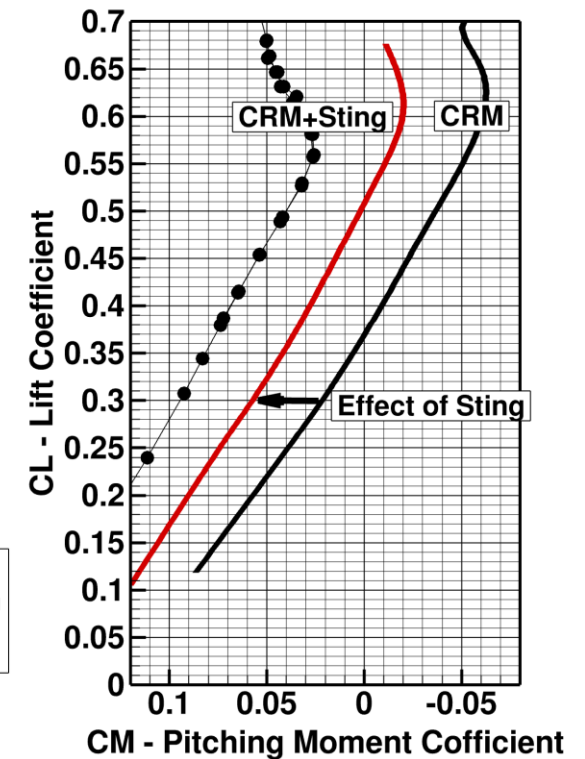
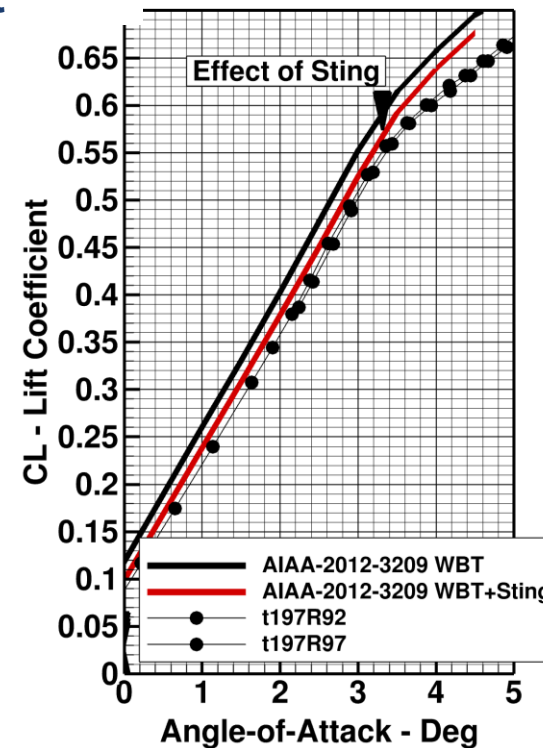
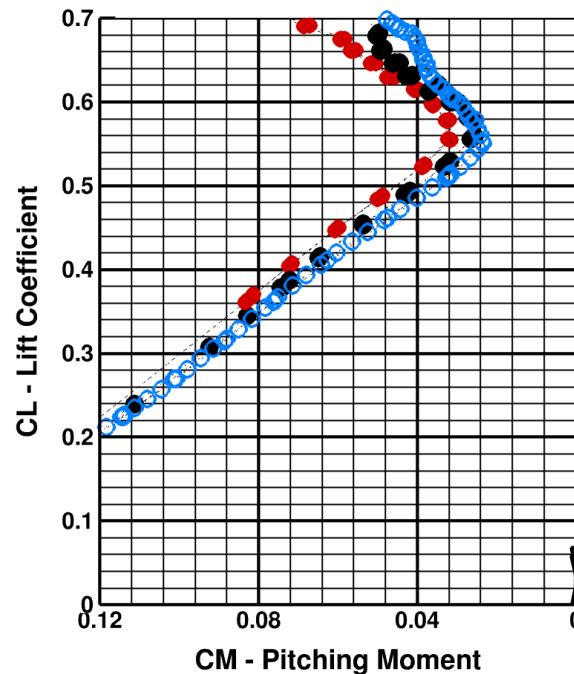
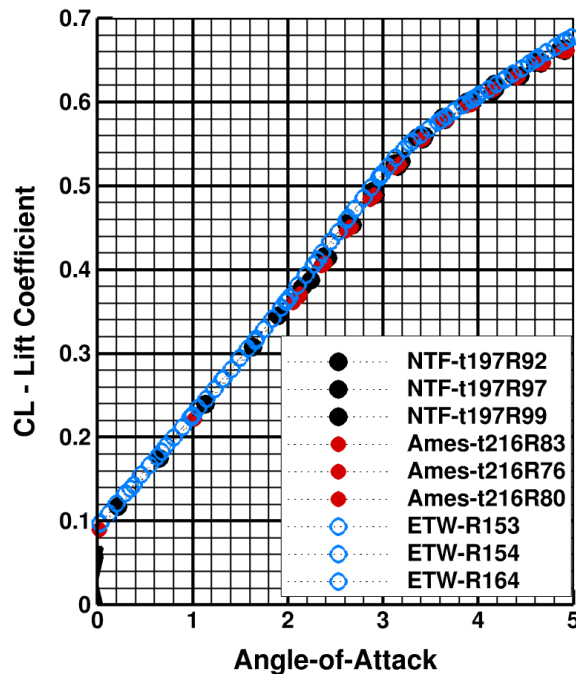
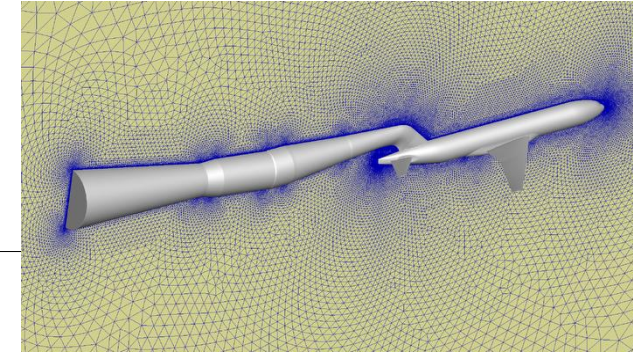
- Test - AOA=4.17°, CL=0.659
- Test - AOA=4.42°, CL=0.673





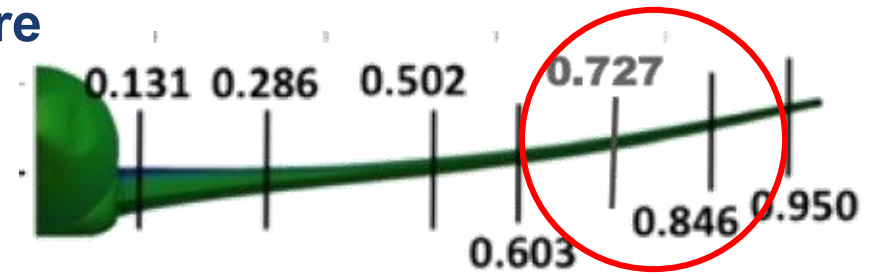
Issue: Solution Offset – Wind Tunnel Environment

- Wind tunnel results from NTF, Ames, and ETW in close agreement
- Different wall system in each tunnel
- Same mounting system used in each tunnel
- Initial CFD reported in AIAA-2012-3209
- Now subject of Working Group #4: Test Environment

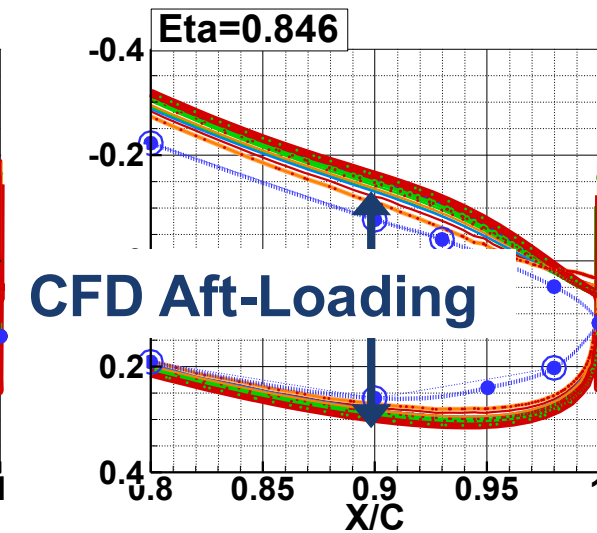
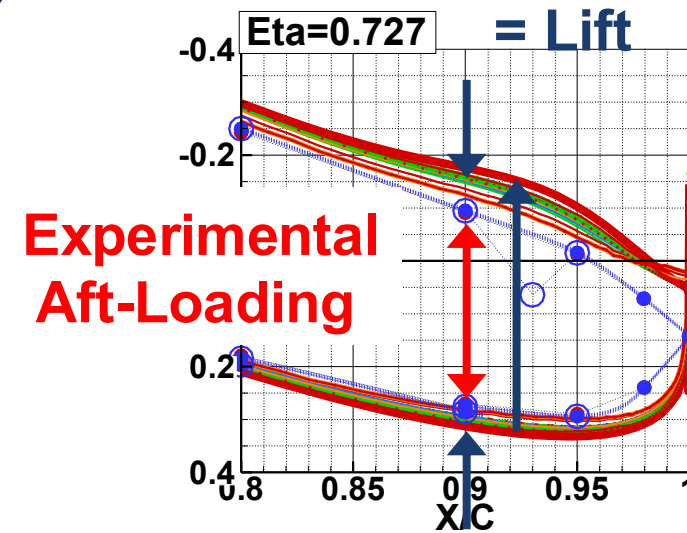
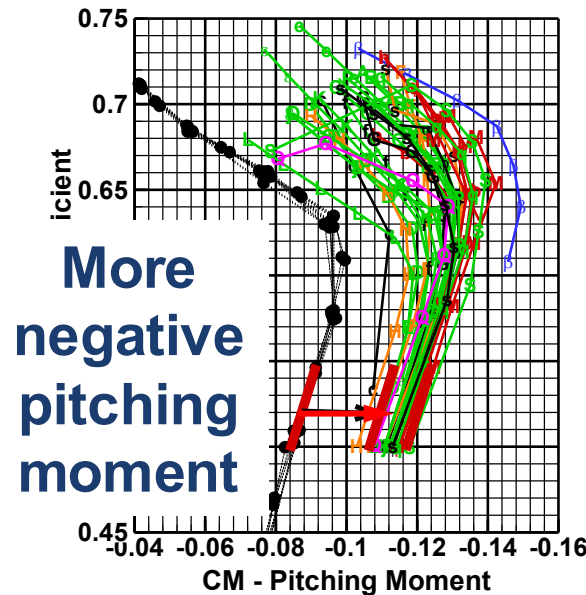
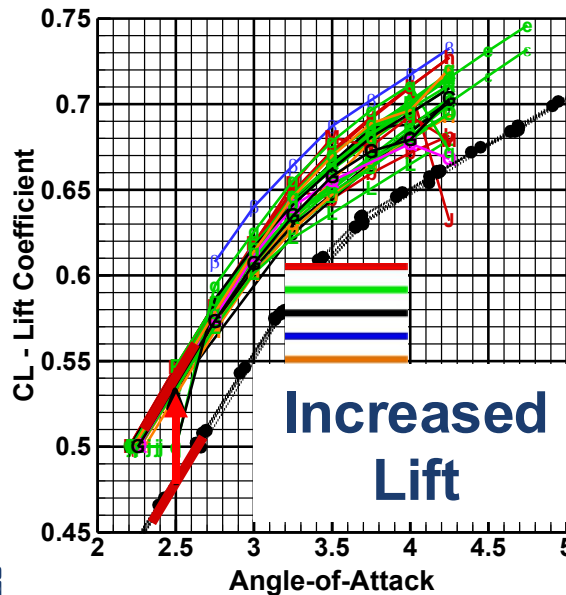


Issue: Solution Offset - Excessive CFD Aft-Loading?

- Excessive aft-loading contributes to greater lift and more negative (nose down) pitching moment
- NOT a geometry problem!
- Seen on other models w/significant aft TE camber - AIAA-2002-0420
- BUT! Appears to account for only a fraction of the off-set
- Wind tunnel mounting system + aft loading??

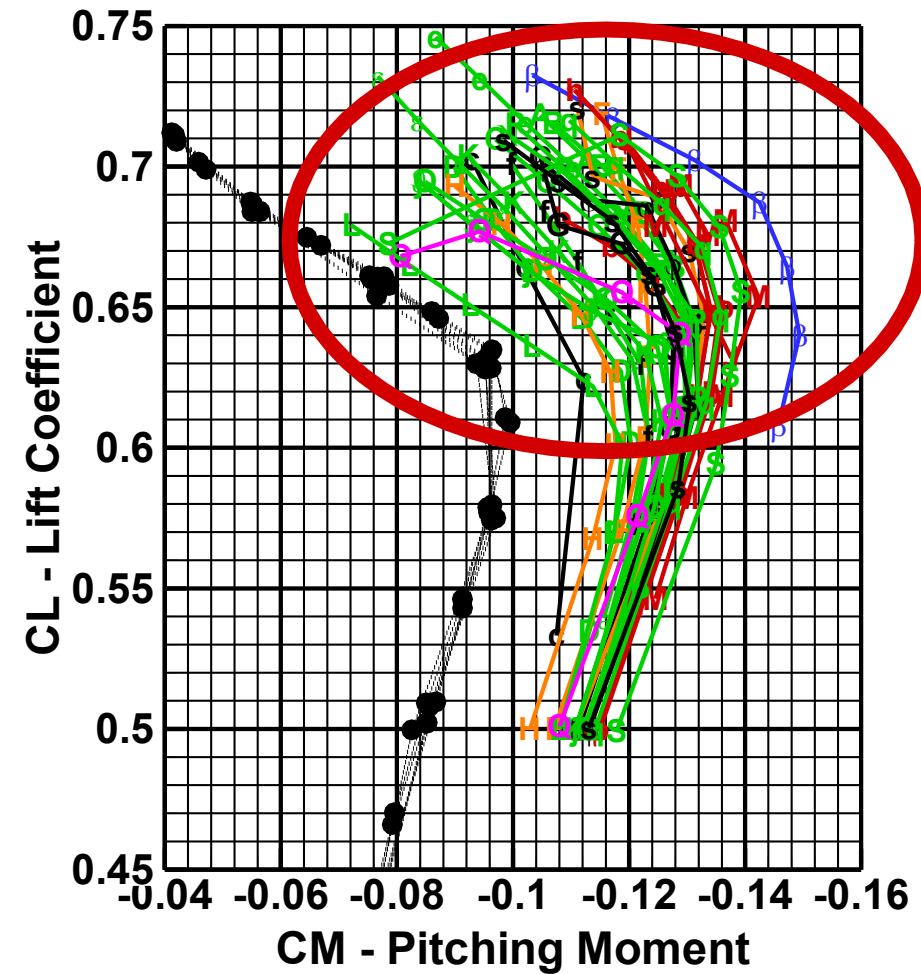
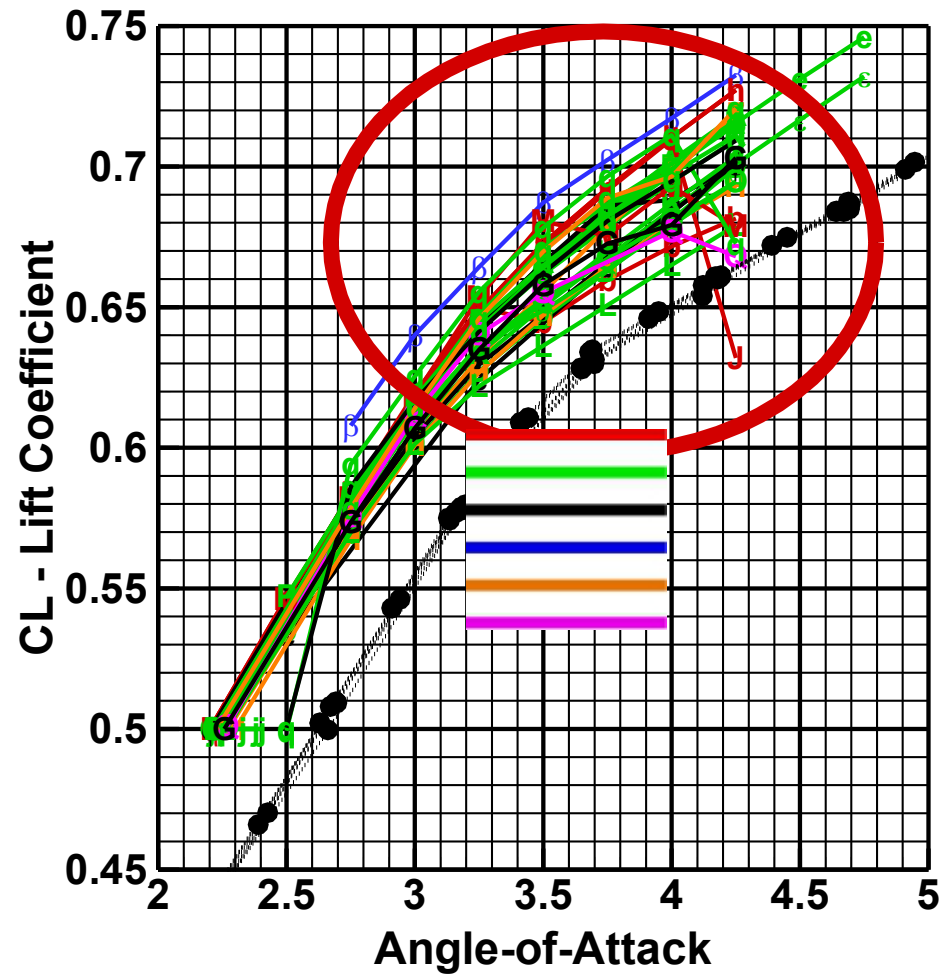


Trailing Edge Pressures CFD Aft-Loading

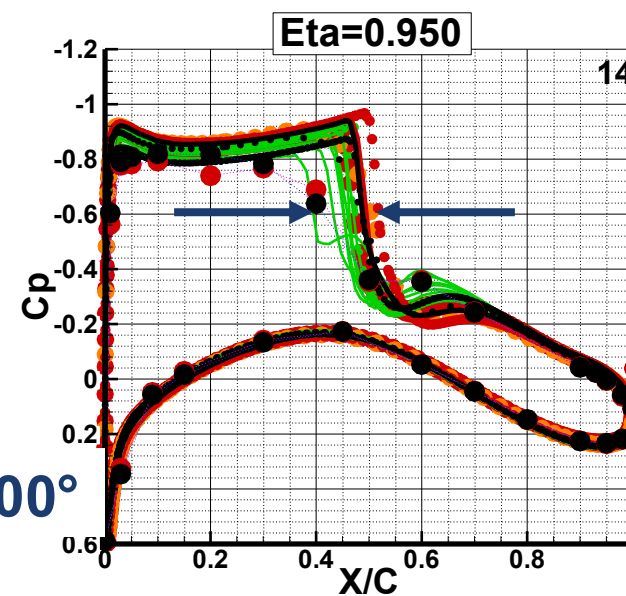
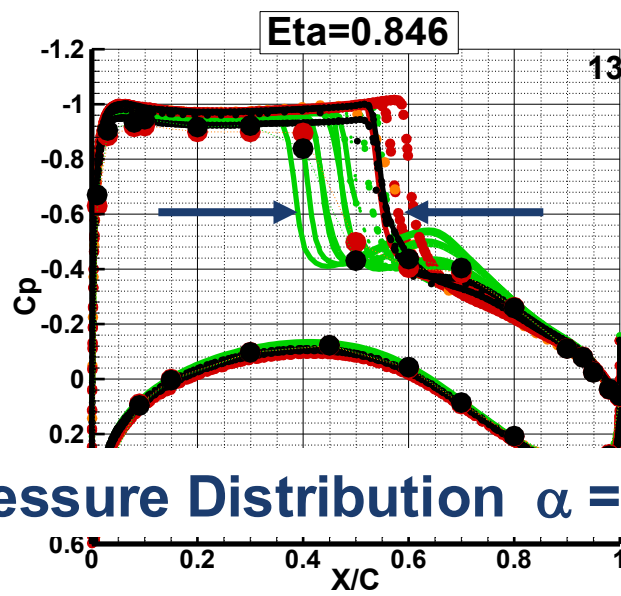
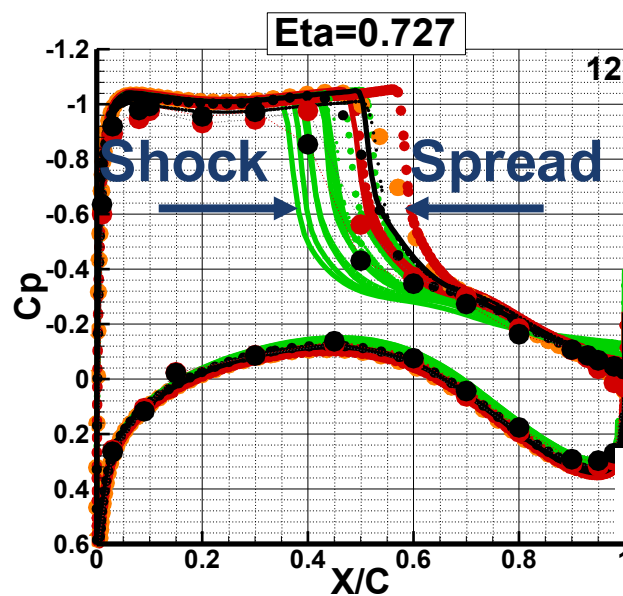
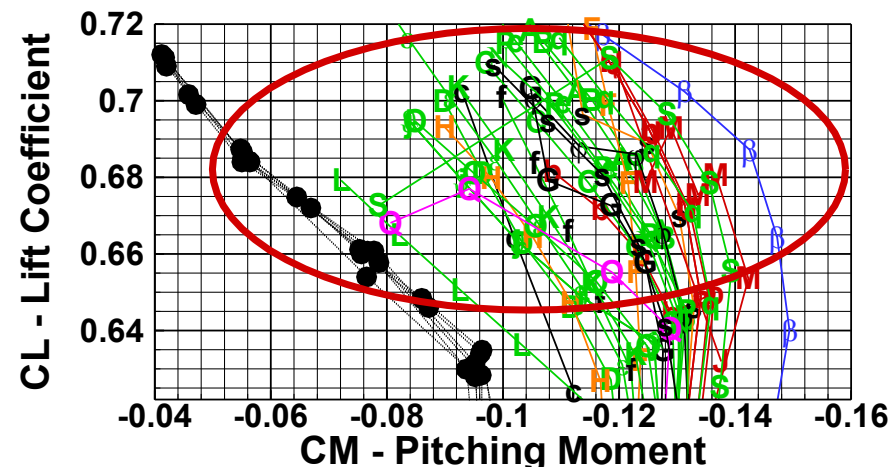
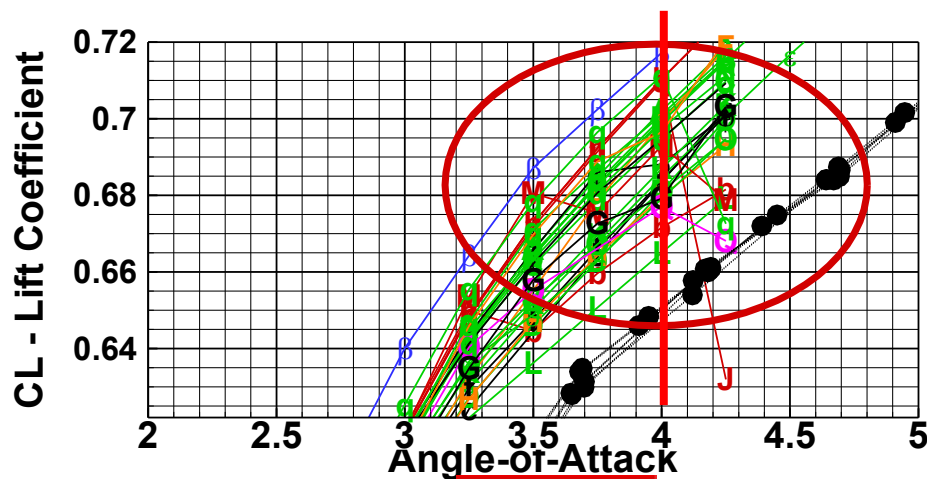


• Experimental Data

Issues: Excessive Solution Spread

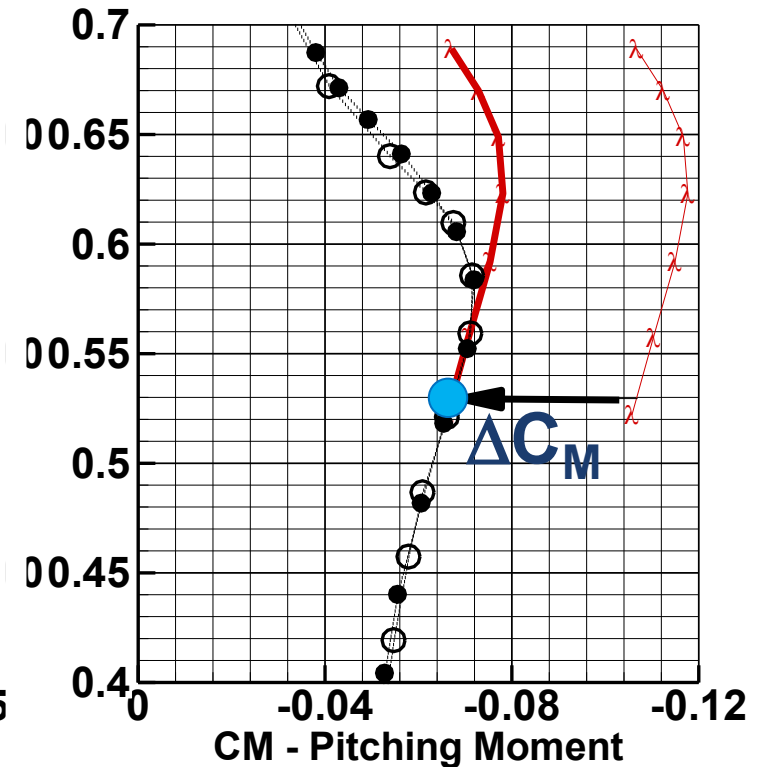
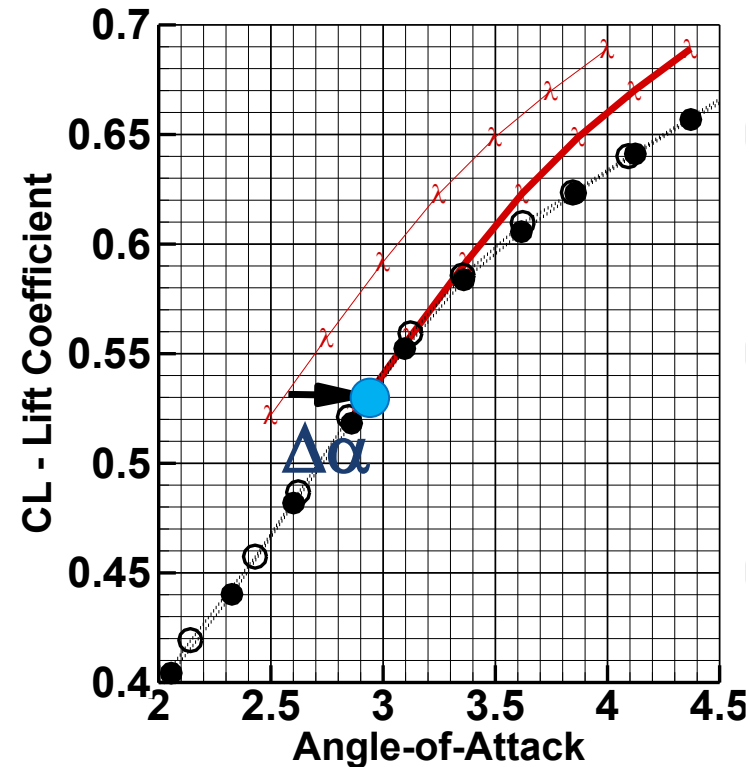


Solution Spread – Predicted Shock Location

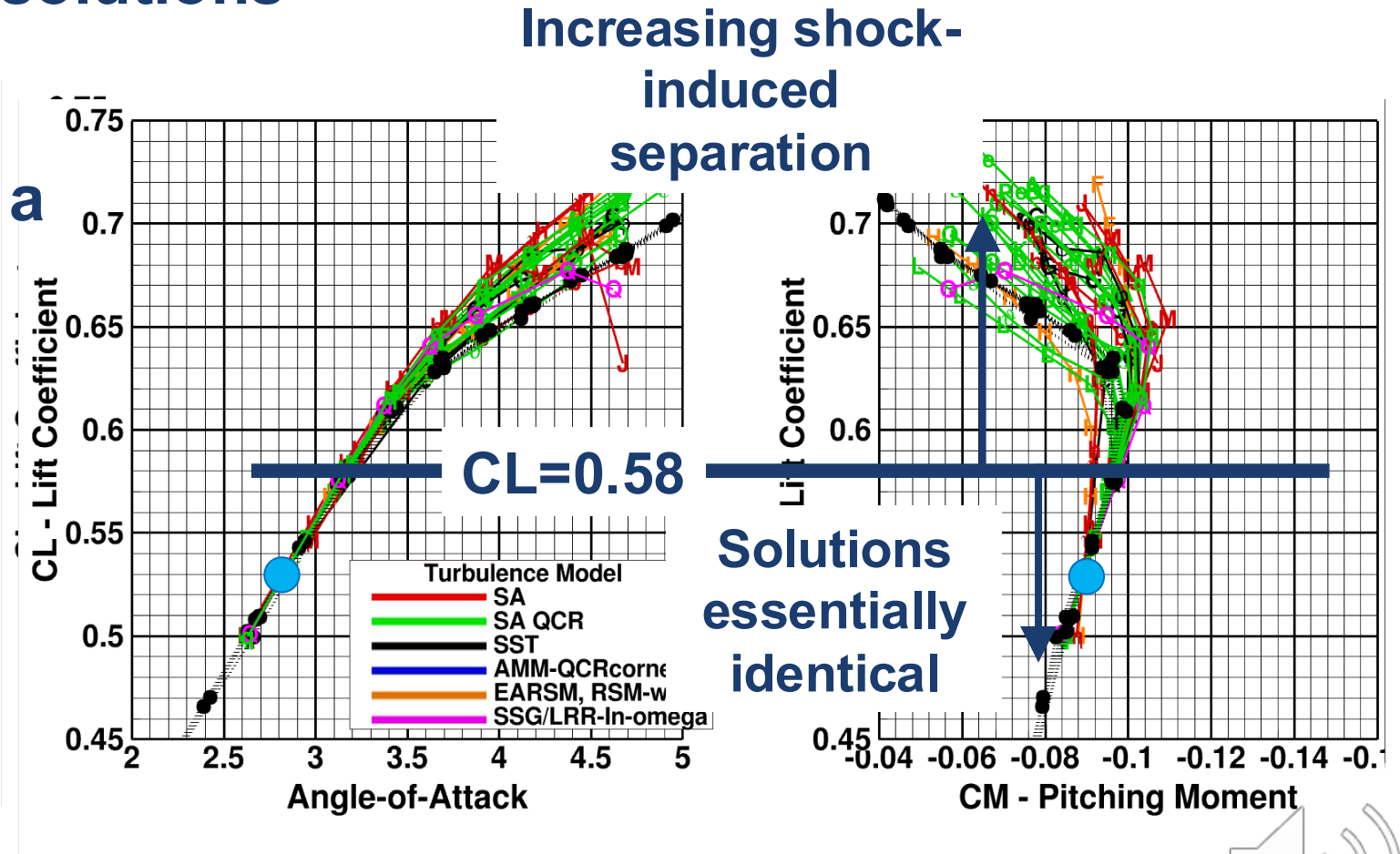


Pressure Distribution $\alpha = 4.00^\circ$

- CFD and WT are better at predicting increments than absolutes.
- Collapse CFD results to pass through a common point by adding a Δ angle-of-attack ($\Delta\alpha$) and Δ pitching moment (ΔC_M) to each solution.
- Clear view of C_L and C_M variation with a variation



- Starting with the original solutions
- Collapse the data to a common point by adding a $\Delta \alpha$ and ΔCM to each solution
- Allows a better look at how the solutions vary with increasing angle-of-attack



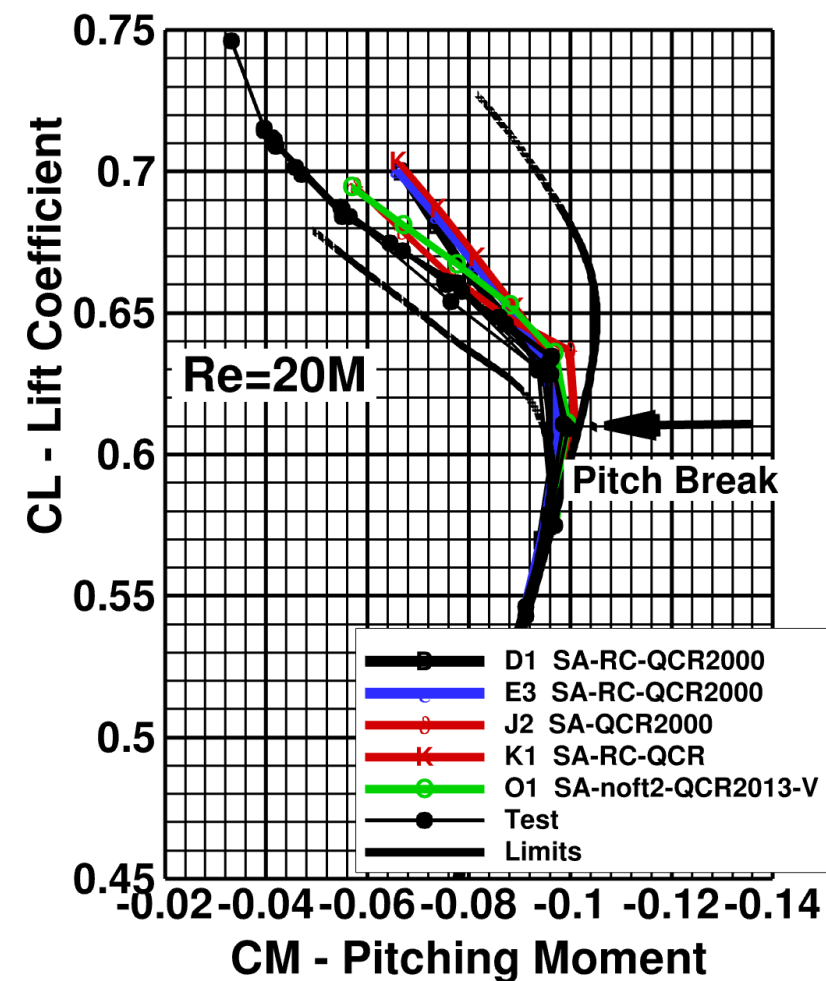
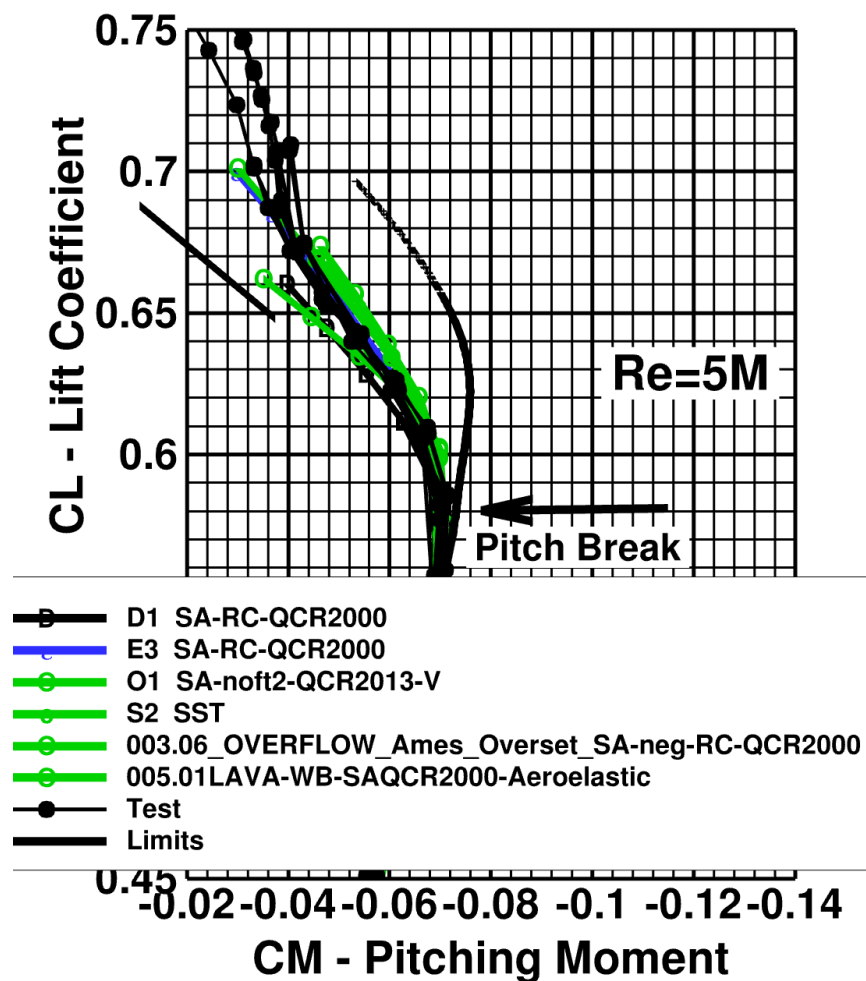
Best Matches

- 6 out of 23 @ $Re = 5M$

Includes DPW8

- 5 out of 31 @ $Re = 20M$

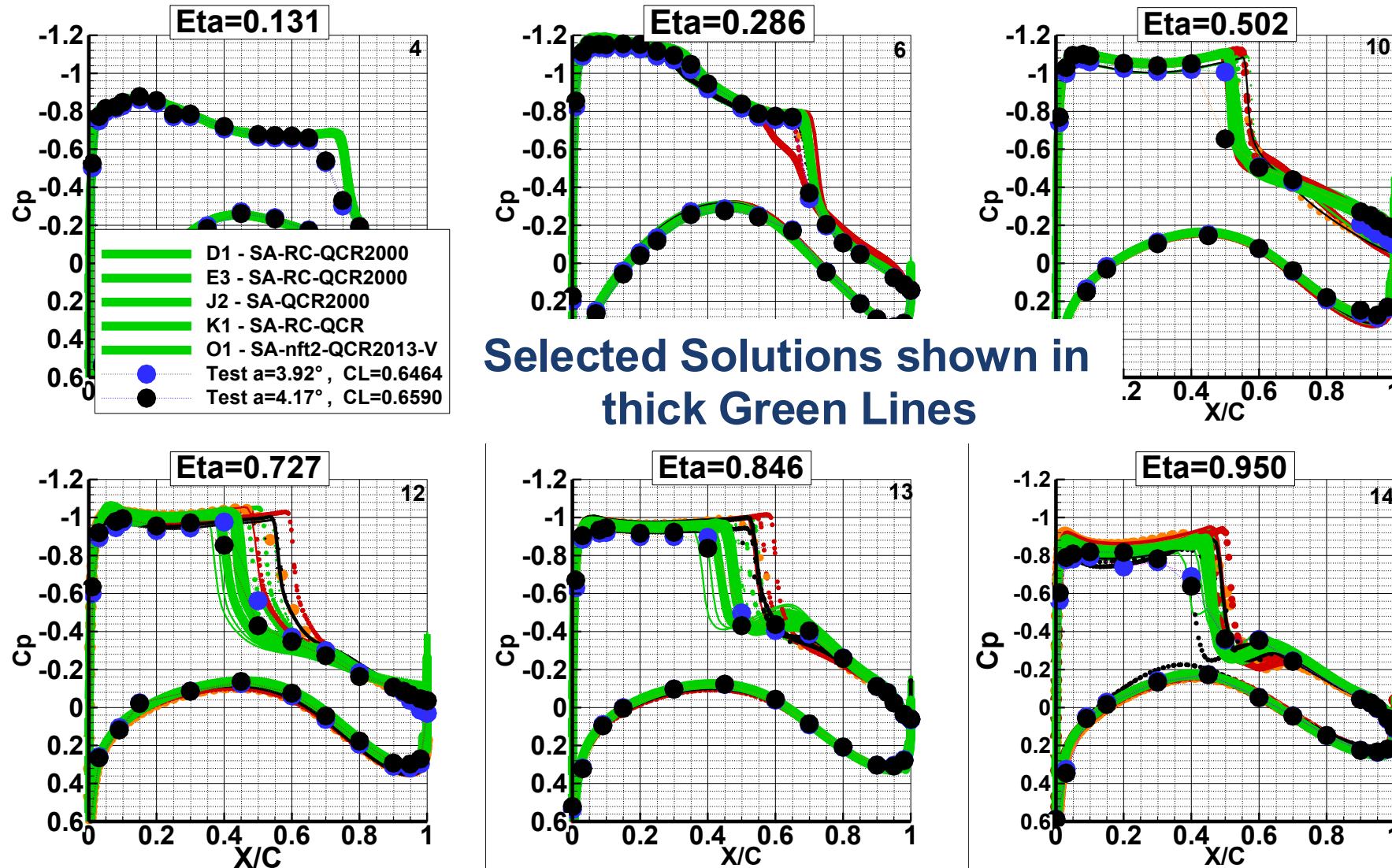
- 3 Solvers best matched both 5M & 20M data



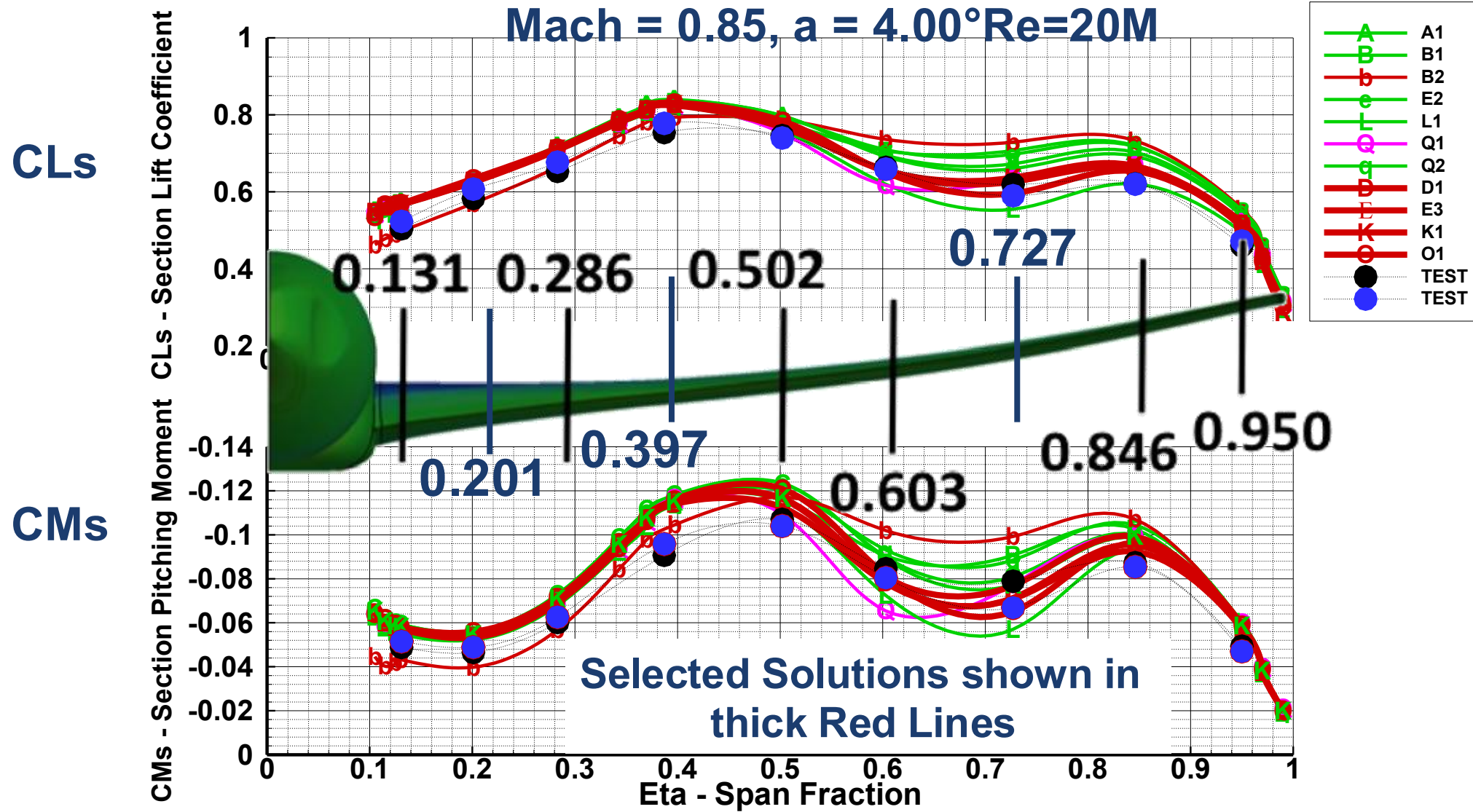
Mach = 0.85, Re=5M, Re=20M-Selected Solutions

- All but one used some variation of SA-QCR turbulence model

Wing Pressure Distributions – 4.0° angle-of-attack



Wing Section Lift and Moment Distribution



- **Attached flow – Drag: Consistent results – Good.**
- **Too much variation of Lift and Pitching moment at higher angles of attack – Shock-induced separation: We need to better understand the interaction of grid, solver, turbulence model.**
- **Need to better understand the issue of the excessive aft loading.**
- **A few solutions matched the test data at the high-angles-of attack very well – but WHY? (Steady RANS vs Unsteady WT test!).**
- **MUST advance the state-of-the-art on solution adaptive grid solvers. Bigger grids are not the answer. Having them in the right place is.**
- **Need further detailed experimental measurements that adequately capture the flow separation and unsteadiness – Could uPSP help?.**

- **DPW-8/AePW-4: Working Groups are focusing on many of these issues. Must continue!**
 - Working Group #1: Source of Scatter Working Group
 - Working Group #2: Static Aeroelastic Deformation Working Group (with AePW)
 - Working Group #3: Buffet Working Group (with AePW)
 - Working Group #4: Wind Tunnel Test Environment Working Group
- **These solution sets and experimental data represent a gold mine of information to further the knowledge of CFD and aerodynamics – GREAT PROJECTS FOR MASTERS STUDENTS.**

Where do we go from here?
DPW-9 ??

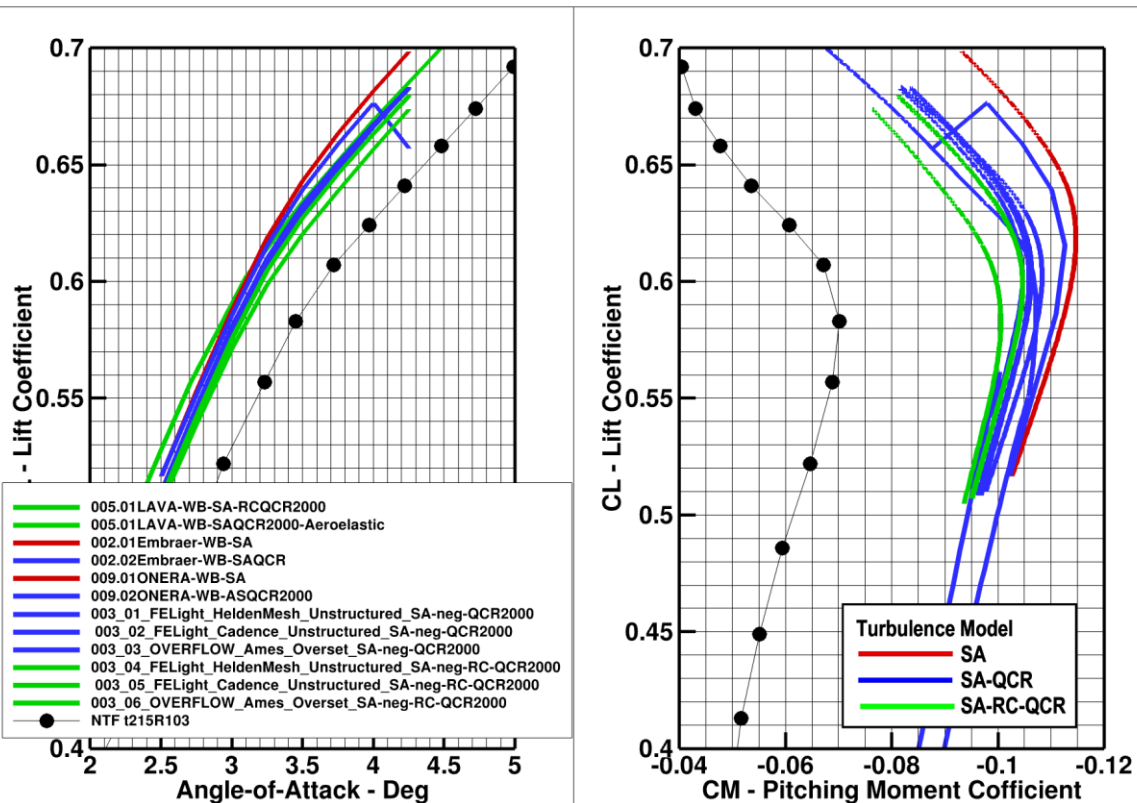


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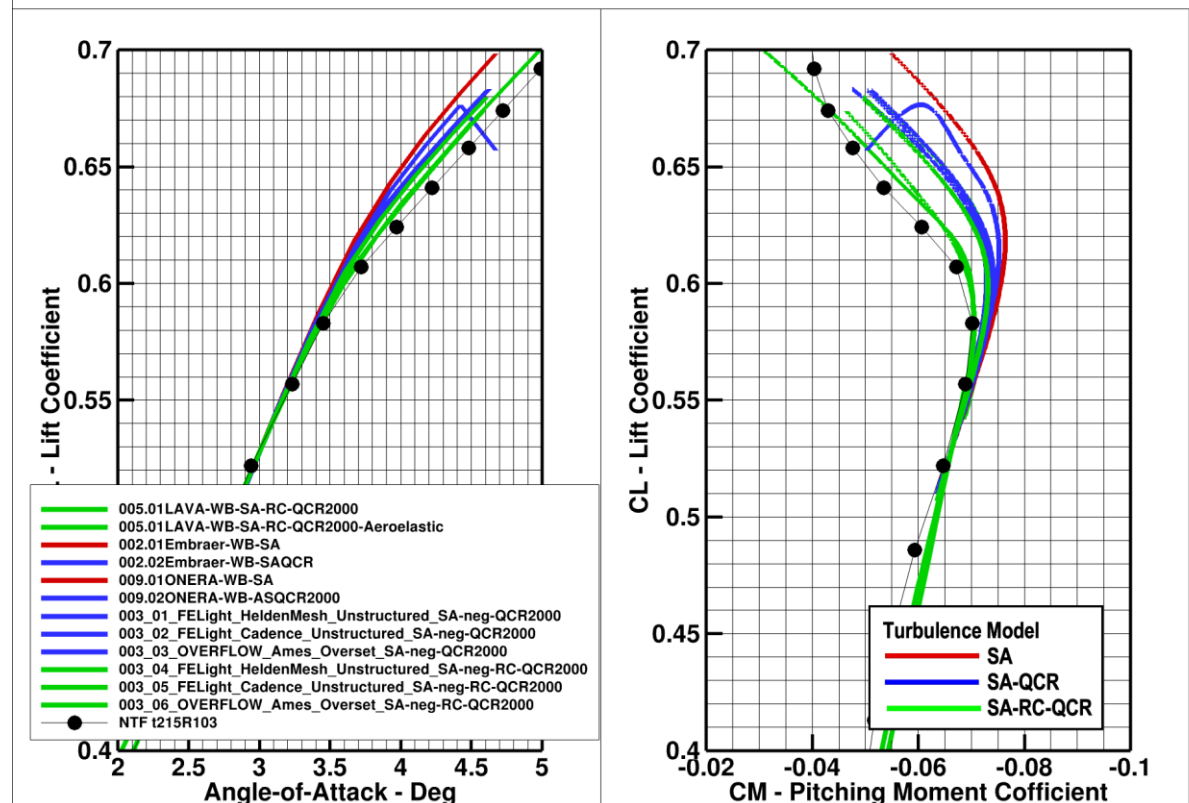
Lift and pitching moment for all solutions – DPW8

Mach = 0.85, Re = 5M

NASA CRM WING-BODY
Mach=0.85, Re=5M



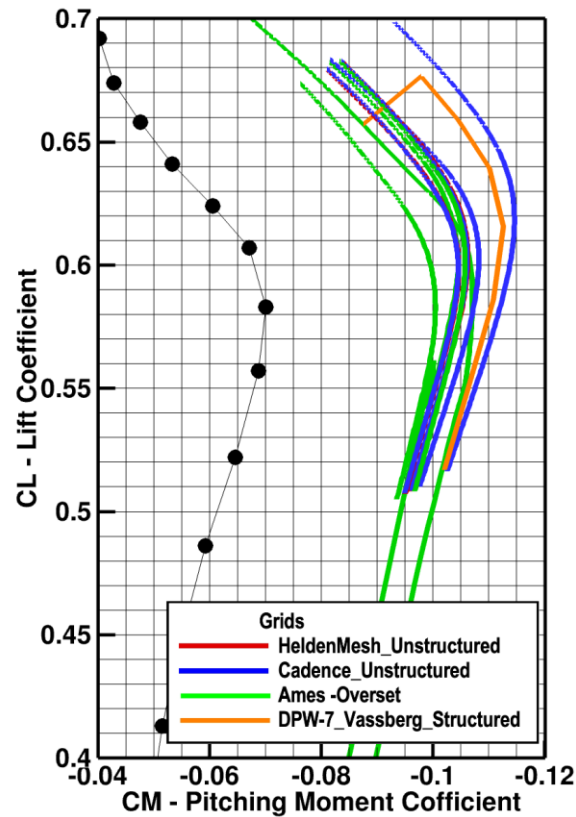
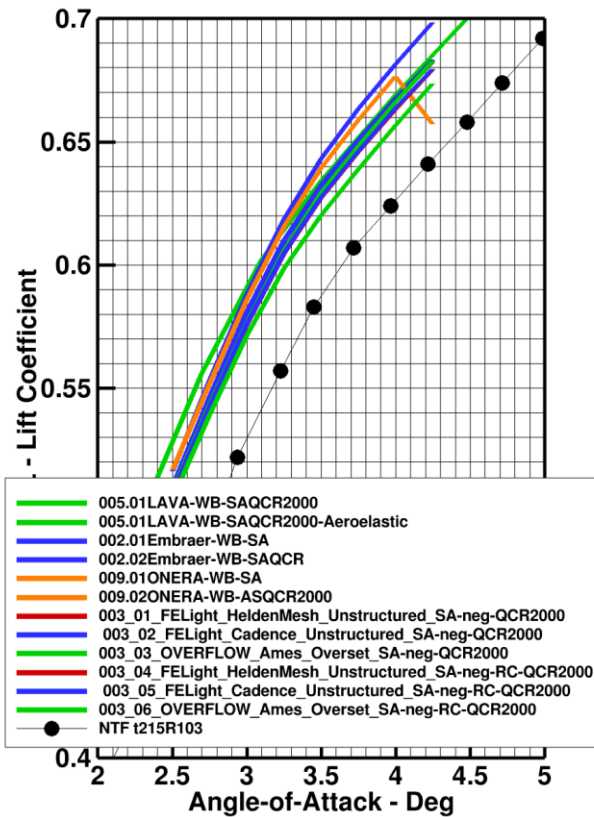
NASA CRM WING-BODY
Mach=0.85, Re=5M



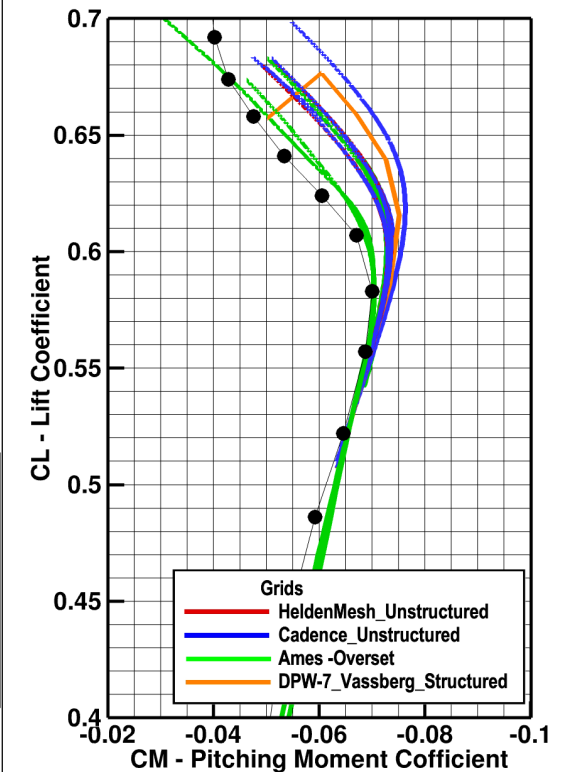
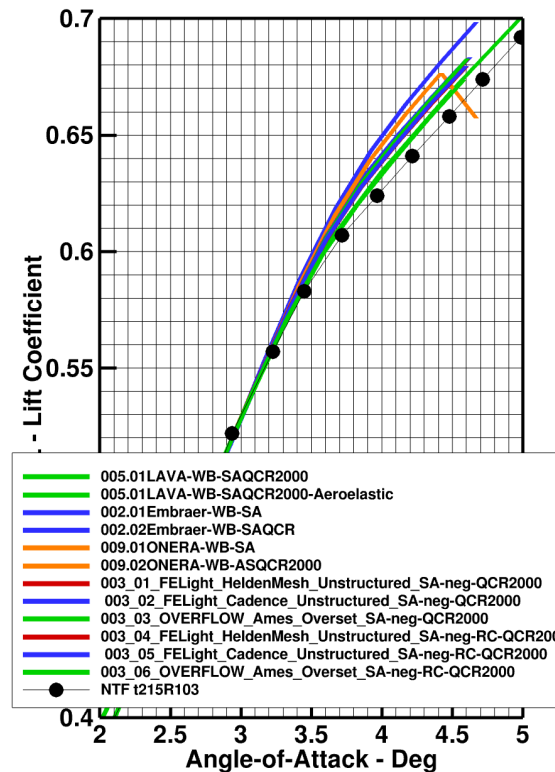
Specific geometry and grids for each α

Mach = 0.85, Re = 5M

NASA CRM WING-BODY
Mach=0.85, Re=5M



NASA CRM WING-BODY
Mach=0.85, Re=5M

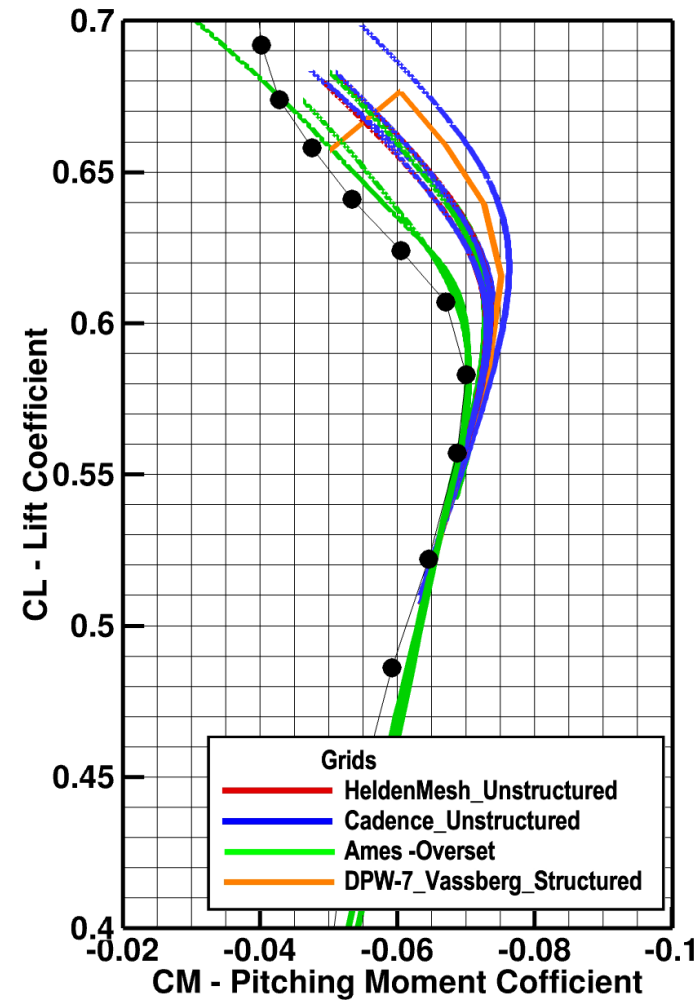
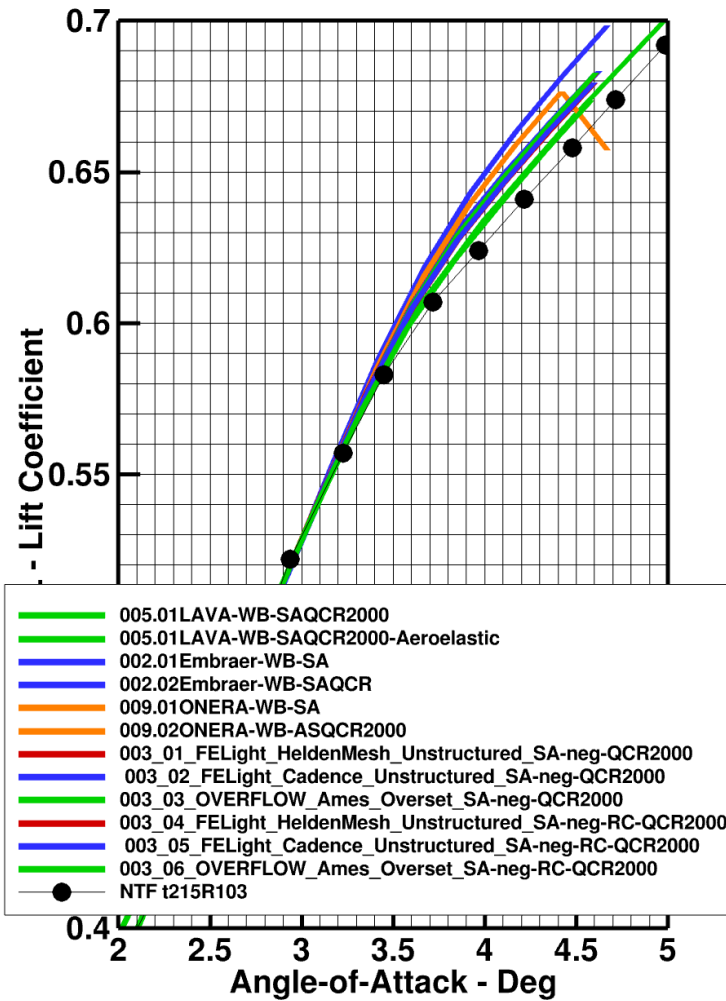


Specific geometry and grids for each α

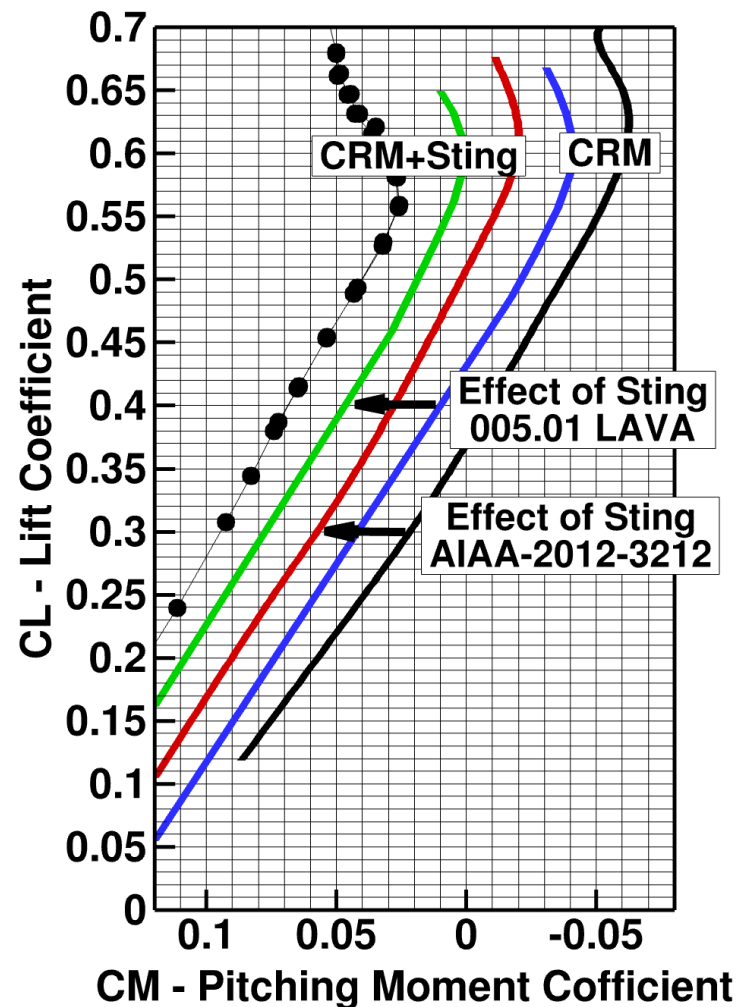
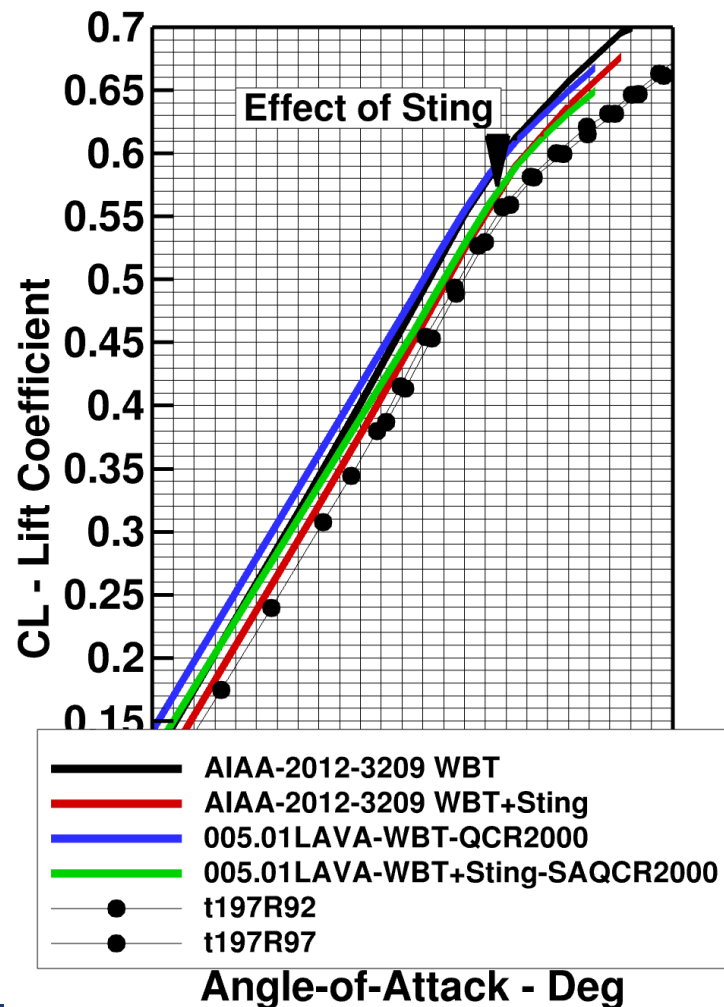
Lift and pitching moment for all solutions

NASA CRM WING-BODY
Mach=0.85, Re=5M

Mach = 0.85, Re = 5M

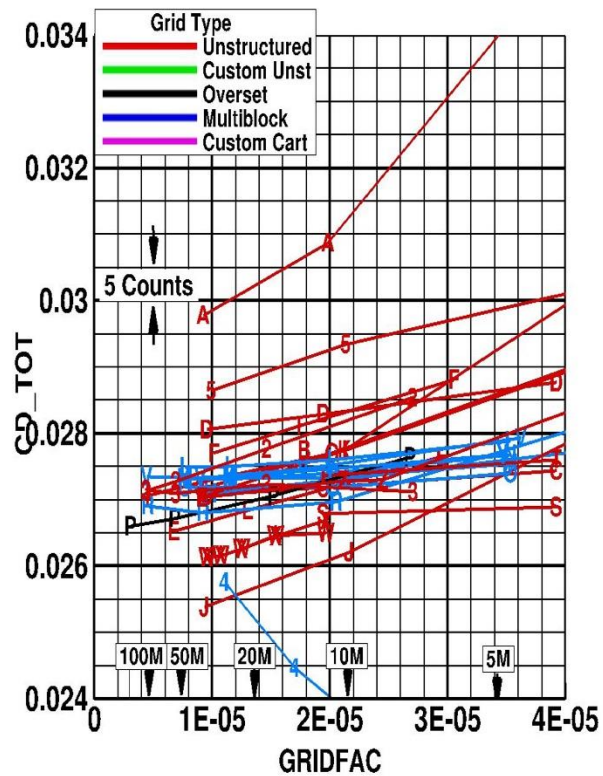


NASA CRM WING-BODY-TAIL
Mach=0.85, Re=5M
Wind Tunnel Mounting System

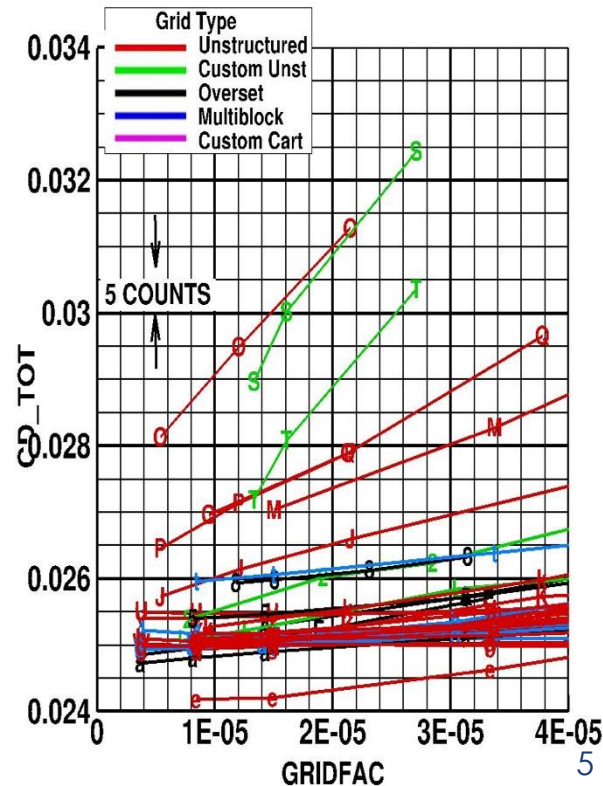


GRID Convergence – Grid Type

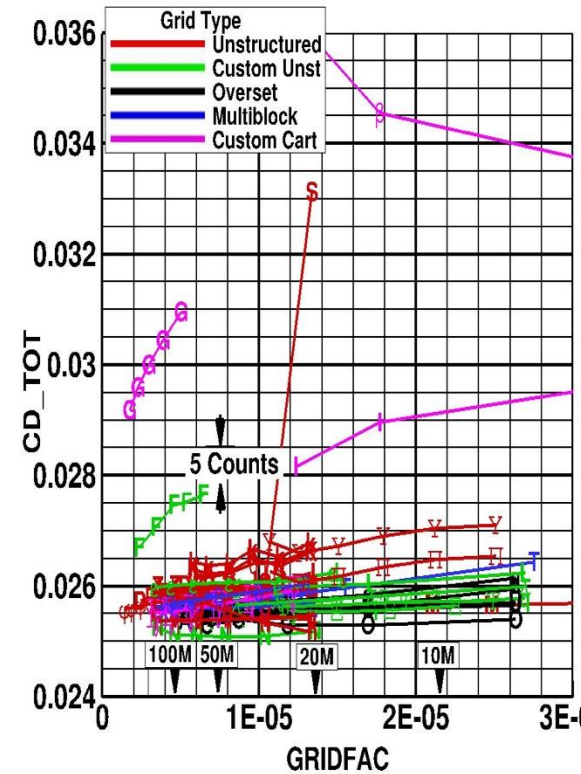
DPW4
Wing-Body-Tail
Re=5M
2009



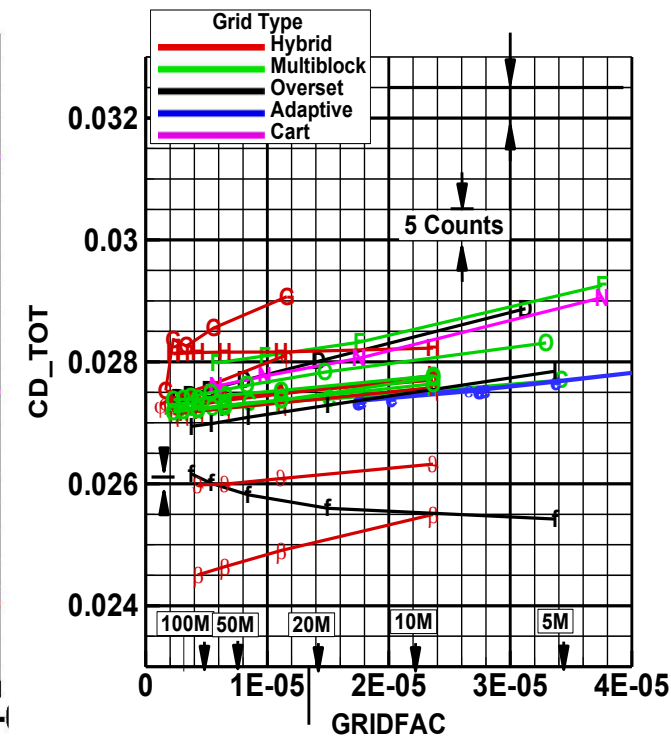
DPW5
Wing-Body
Re=5M
2012



DPW6
Wing-Body
Re=5M
2016



DPW7
Wing-Body
Re=20M
2022



Drag Polar for all solutions

